THE ERDŐS–STRAUS EQUATION ON EXPONENTIAL FAMILIES BEYOND EVERY ELSHOLTZ–TAO COVERING

ANONYMOUS

Abstract. For an exponential family $n_s = c_0\,2^s + k$ the residues of n_s modulo any fixed integer are eventually periodic in s, so one may try to prove the Erdős–Straus conjecture $4/n = 1/x + 1/y + 1/z$ for all members at once by a finite covering of the exponents with Elsholtz–Tao identity classes and divisor classes, in the manner of Sierpiński's coverings for $k \cdot 2^n + 1$. When $c_0 + k = 4^J$ and $k \equiv 1 \pmod 8$ this is impossible: along $s \equiv 0 \pmod L$ the members imitate the even square 4^J modulo every odd number, and an odd-square (Schinzel–Yamamoto) argument shows that no finite set of such classes covers them. We show that these obstructed families can nevertheless be settled completely, by adding a few identity classes whose coordinates are polynomials in a root $Y = 2^{s/D}$ of the exponential rather than in n. Their divisors are forced by an algebraic factorisation along the family (Sophie Germain's identity $4X^4 + 1 = (2X^2 + 2X + 1)(2X^2 - 2X + 1)$, cyclotomic factors of $Y^h + w^h$, or a binomial $n + 4a^2d$), in the spirit of Izotov's construction for Sierpiński numbers. We prove the Erdős–Straus conjecture for every $n = 3 \cdot 2^s + 1$, every $63 \cdot 2^s + 1$ and every $7 \cdot 4^s + 9$ (all three provably beyond every finite Elsholtz–Tao covering), and for every $4^s + k$ with $k \in \{9, 57, 81, 105, 129, 153, 177, 201, 225, 249\}$; for $15 \cdot 2^s + 1$ we prove it for every $s \not\equiv 0 \pmod{60}$. All proofs are explicit identity certificates; most were found by computer search and are re-verified by a short self-contained script. On the negative side we prove that no identity class of constant gap $4x - n$ or constant ratio x/n, polynomial in any root $2^{s/D}$, reaches the unit-square point of $15 \cdot 2^s + 1$, $31 \cdot 4^s + 33$, $7 \cdot 4^s + 57$, $63 \cdot 4^s + 1$, $47 \cdot 4^s + 17$ or $55 \cdot 4^s + 9$, and we close one branch of the moving-gap binomial menu at $15 \cdot 2^s + 1$. The families are thin; nothing here bears on the Erdős–Straus conjecture for primes in general.

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1. Introduction

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers. If $n = ab$ and (1) is solvable for a, it is solvable for n (multiply x, y, z by b), so it suffices to treat primes. The conjecture has been verified numerically up to 10^{17} by Salez [10] and up to 10^{18} by Mihnea and Dumitru [6].

Identity classes and their limits. The classical tool is the polynomial identity on a residue class. Mordell [7] showed that every n outside the six classes $1, 121, 169, 289, 361, 529 \pmod{840}$ (the squares modulo 840) is covered by such identities, and Elsholtz and Tao [2, Prop. 1.9] essentially classified the residue classes that are *solvable by polynomials*: up to finitely many primes they are unions of classes of seven explicit families (Table 1). A primitive class whose residue is a perfect square is never solvable by polynomials (Mordell, Schinzel [7, 11]; it also follows from the vanishing of Type I and Type II solutions at odd squares, [2, Prop. 1.6]). Elsholtz and Tao remark that this "rules out … a finite set of covering congruence strategies" for ES [2, p. 6]. A companion note [14] studies this obstruction for sets of primes from the profinite point of view.

Exponential families. Here we look at thin families

$$\mathcal{F}(c_0, b, k) = \{\, n_s = c_0\, b^s + k : s \geq 0 \,\}, \qquad b \in \{2, 4\},$$

with c_0, k positive integers. The residue of n_s modulo a fixed M is eventually periodic in s, so a finite list of identity classes may cover *all* exponents, exactly as Sierpiński [12] proved that $k \cdot 2^n + 1$ is composite for all n by covering the exponents n with finitely many congruences (see [3, §1]). We call such a list an *exponent covering*. For many families it exists and is small (Theorem 4.11); the companion work [15] certified, for instance, ES for every $315 \cdot 2^m + 1$ and every $15015 \cdot 2^m + 1$.

The obstruction is a *unit-square point*. If c_0 is odd, $k \equiv 1 \pmod 8$ and $c_0 + k = 4^J$, then for $s \equiv 0 \pmod L$ with L highly divisible, $n_s \equiv c_0 + k = 4^J$ modulo every odd number of the form $2^L - 1$ and $n_s \equiv k \pmod{2^s}$; so n_s is congruent to an odd perfect square modulo every fixed modulus, and the odd-square argument shows that no finite set of Elsholtz–Tao classes and

divisor classes covers the family (Theorem 2.7). Examples are $3 \cdot 2^s + 1$ (the Proth numbers with $k = 3$), $15 \cdot 2^s + 1$, $63 \cdot 2^s + 1$, $7 \cdot 4^s + 9$, $31 \cdot 4^s + 33$.

Results. The point of this note is that the obstruction concerns identities that are polynomial in n, not ES itself. Along a residue class of exponents the integer n_s is a polynomial in a root $Y = 2^{(s-s_0)/D}$, and an identity polynomial in Y is again valid on a whole residue class of s. Such an identity can use divisors that move with n and are forced by an algebraic factorisation of Y-polynomials, and these can reach the unit-square point.

Main Theorem. (a) (Theorems 4.1, 4.4, 4.6) *The Erdős–Straus conjecture holds for every $n = 3 \cdot 2^s + 1$, every $n = 63 \cdot 2^s + 1$ and every $n = 7 \cdot 4^s + 9$ ($s \geq 0$). None of these families admits a finite covering by Elsholtz–Tao classes and divisor classes (Theorem 2.7).*

(b) (Theorems 4.9, 4.11) *It holds for every $n = 4^s + k$, $s \geq 0$, for $k \in \{9, 57, 81, 105, 129, 153, 177, 201, 225, 249\}$.*

(c) (Proposition 5.12) *It holds for every $n = 15 \cdot 2^s + 1$ with $60 \nmid s$.*

(d) (Theorem 5.7, Corollary 5.9) *For $(c_0, k) \in \{(15, 1), (31, 33), (7, 57), (63, 1), (47, 17), (55, 9)\}$ no identity family that is polynomial in a root $Y = 2^{s/D}$ (any D) and has constant gap $4x - n$ or constant ratio x/n is valid on a neighbourhood $\{s \equiv 0 \pmod{L},\ s \geq s_0\}$ of the unit-square point. Hence these families have no finite covering by Elsholtz–Tao classes, divisor classes and such families.*

Parts (a)–(c) are PROVED; the certificates of (a) for $63 \cdot 2^s + 1$ and $7 \cdot 4^s + 9$, of (b) and of (c) were found by computer search and the proofs are *computer-assisted* in the sense that a finite coverage statement (a list of at most 9180 residues) is checked by machine; every identity is printed in the text and can be checked by hand. The proof for $3 \cdot 2^s + 1$ (Theorem 4.1) is short and fully hand-checkable. Part (d) is PROVED up to a finite enumeration (Computation 5.8). In §5.2 we also prove that one of the two branches of the moving-gap menu that produced (a) for $63 \cdot 2^s + 1$ is empty for $15 \cdot 2^s + 1$ (Theorem 5.15); the other branch, and ES for $15 \cdot 2^s + 1$ with $60 \mid s$, remain open.

The mechanism, and a precedent. In all three families of (a) the unit-square point is reached by a single kind of move. For $3 \cdot 2^s + 1$ with $4 \mid s - 4$ put $X = 2^{(s-4)/4}$, so $n = 48X^4 + 1$ and $x := (n + 11)/4 = 3(4X^4 + 1)$. Sophie Germain's identity

$$4X^4 + 1 = (2X^2 + 2X + 1)(2X^2 - 2X + 1)$$

supplies divisors of x that are not powers of fixed primes, and the elementary window lemma (Lemma 3.1) turns a congruence between them modulo the *window* $m = 4x - n = 11$ into a solution of (1). The congruence holds on whole residue classes of s, including a neighbourhood of the unit-square point. This is the ES counterpart of Izotov's observation [4] (see [3, pp. 3–4]) that a Sierpiński-type covering may omit one congruence class of exponents if the family factors algebraically there, by the same factorisation $4x^4 + 1 = (2x^2 + 2x + 1)(2x^2 - 2x + 1)$. For $4^s + 105$ the closing class uses $u^3 + 1 = (u + 1)(u^2 - u + 1)$, $u = 4^{(s-4)/3}$; for $63 \cdot 2^s + 1$ it is a Type II family with a *moving* gap $4x - n = 3(Y^2 - 3Y + 9)$ built from $Y^3 + 27 = (Y + 3)(Y^2 - 3Y + 9)$ (§3). Part (d) shows that the constant-gap version of the mechanism has a precise 2-adic limit: it can reach a unit-square point only through a Sophie Germain piece whose value at the limit has odd 2-adic valuation (Lemma 5.3), and an explicit finite check shows that no such piece exists for the six listed families.

Scope. Each family has $O(\log X)$ members up to X, so these theorems say nothing about ES for primes in general; a finite set of thin identity classes misses almost all primes of any residue class (see [19, Prop. 5.1]). The members below 10^{18} (for instance $s \leq 53$ for $63 \cdot 2^s + 1$, $s \leq 29$

for $4^s + k$) are already covered by the numerical verifications [10, 6]. Each family lies in a finite union of residue classes, and several of our families are settled by Mordell's identities for most s; we say where the new content lies. We found no prior work on ES for exponential families (searches of zbMATH Open for all documents on the Erdős–Straus equation, and web searches; see [24]); this is not a claim of priority.

Labels. PROVED means that a complete proof is given here. COMPUTER-ASSISTED means that a finite verification (always of a statement about finitely many explicit residues and identities) is done by the script `scripts/c1_expfam_verify.py` (Appendix A), which uses only exact arithmetic and imports nothing from the search code; the original certificates (all except that of Proposition 5.12, which is new here) were in addition checked by two independent reviewers' programs each. EVIDENCE means a computation that supports but does not prove a statement. OPEN and CONJECTURE are not results.

Provenance. This note consolidates internally reviewed work of a research notebook: [15] (exponent coverings, the unit-square criterion), [16] ($4^s + k$), [18] ($4^s + 105$), [19] ($3 \cdot 2^s + 1$), [20] ($7 \cdot 4^s + 9$ and a second proof for $3 \cdot 2^m + 1$), [22] ($63 \cdot 2^s + 1$ and Theorem 5.7) and [23] ($15 \cdot 2^s + 1$). Several proofs are re-derived and expanded here; Proposition 5.12 is new in this note.

2. Exponent coverings and unit-square points

2.1. **The Elsholtz–Tao classes.** We use the seven families of [2, Prop. 1.9] in the notation of [14, §2]. Each is indexed by a triple κ of positive integers subject to a side condition and defines a set of residue classes modulo $M(\kappa)$ (Table 1). On its class each family carries an explicit solution

$$\pi^I(a,b,c,d) = (abdn,\ acd,\ bcd) \quad \text{(I1–I4)}, \qquad \pi^{II}(a,b,c,d) = (abd,\ acdn,\ bcdn) \quad \text{(II1–II3)},$$

where two or three of a, b, c, d depend on κ alone, and the remaining ones are the polynomials in n listed in the table.

family	κ	side condition	class	remaining coordinates
I1	a, d, f	$f \mid 4a^2d + 1$	$n \equiv -f \pmod{4ad}$	$e = \frac{4a^2d+1}{f}$, $b = \frac{ne+1}{4ad}$, $c = \frac{n+f}{4ad}$
I2	a, c, f	$(4ac, f) = 1$	$n \equiv -f\ (4ac)$, $n \equiv -c/a\ (f)$	$d = \frac{n+f}{4ac}$, $b = \frac{na+c}{f}$
I3	c, d, f	$(4cd, f) = 1$	$n \equiv -f\ (4cd)$, $f \mid n^2 + 4c^2d$	$a = \frac{n+f}{4cd}$, $b = \frac{n^2+4c^2d+nf}{4cdf}$
I4	a, b, e	$e \mid a + b$, $(e, 4ab) = 1$	$n \equiv -1/e \pmod{4ab}$	$c = \frac{a+b}{e}$, $d = \frac{ne+1}{4ab}$
II1	a, b, e	$e \mid a + b$, $(e, 4ab) = 1$	$n \equiv -e \pmod{4ab}$	$c = \frac{a+b}{e}$, $d = \frac{n+e}{4ab}$
II2	a, d, f	$4ad \mid f + 1$	$n \equiv -4a^2d \pmod{f}$	$c = \frac{f+1}{4ad}$, $b = \frac{nc+a}{f}$
II3	a, d, e	$(4ad, e) = 1$	$n \equiv -4a^2d - e \pmod{4ade}$	$c = \frac{n+4a^2d+e}{4ade}$, $b = \frac{n+e}{4ad}$

TABLE 1. The Elsholtz–Tao families [2, Prop. 1.9].

Lemma 2.1 (PROVED; [2, §§2,10], [14, Lemma 2.1])**.** *If κ is admissible and the integer $n \geq 1$ lies in its class, the coordinates of Table 1 are positive integers and π^I (resp. π^{II}) is a solution of (1). Moreover $4/n = 1/(abdn) + 1/(acd) + 1/(bcd)$ is equivalent to $4abcd = n(a+b) + c$, and $4/n = 1/(abd) + 1/(acdn) + 1/(bcdn)$ to $4abcd = nc + a + b$.*

The last two equivalences are immediate on clearing denominators. We call the class of κ an *ET class*. A *divisor class* is $D_q = \{n \equiv 0 \pmod q\}$ for an odd prime q; on it $(x,y,z) = \frac{n}{q}(x_q, y_q, z_q)$ for a fixed solution (x_q, y_q, z_q) of $4/q = 1/x + 1/y + 1/z$. (All divisor classes used below have $q \le 541$, and the script records explicit solutions.)

2.2. **Odd squares.** The following is the classical quadratic-reciprocity obstruction of Schinzel and Yamamoto [13], in the form proved by Elsholtz and Tao for Type I [2, §4] (they omit the Type II case). We give both cases, since every negative statement below rests on it.

Lemma 2.2 (PROVED)**.** *Let n be an odd perfect square.*

(i) *There are no positive integers a,b,c,d with $4abcd = n(a+b)+c$ and $\gcd(c,n)=1$.*
(ii) *There are no positive integers a,b,c,d with $4abcd = nc+a+b$ and $\gcd(abd,n)=1$.*

Proof. Note $n \equiv 1 \pmod 8$, and write q for the odd part of ab. We use the Jacobi symbol and the reciprocity law for odd positive coprime q, e: $\left(\frac{q}{e}\right) = \left(\frac{e}{q}\right)(-1)^{\frac{q-1}{2}\frac{e-1}{2}}$.

(i) Since $c \mid n(a+b)$ and $\gcd(c,n)=1$, $e := (a+b)/c$ is a positive integer, and $4abd = ne+1$. Hence $\gcd(e, ab) = 1$, $\gcd(n,q) = 1$, and $ne \equiv -1 \pmod 4$, so $e \equiv 3 \pmod 4$. Modulo q, $e \equiv -n^{-1}$, and n is a square, so $\left(\frac{e}{q}\right) = \left(\frac{-1}{q}\right)$. By reciprocity with $e \equiv 3 \pmod 4$, $\left(\frac{q}{e}\right) = \left(\frac{e}{q}\right)\left(\frac{-1}{q}\right) = 1$. On the other hand $b \equiv -a \pmod e$ and $\gcd(a,e) = 1$, so $\left(\frac{ab}{e}\right) = \left(\frac{-a^2}{e}\right) = \left(\frac{-1}{e}\right) = -1$. Writing $ab = 2^v q$ we get $\left(\frac{2}{e}\right)^v = -1$, so $v \ge 1$. Then $8 \mid 4abd = ne+1$, so $e \equiv 7 \pmod 8$ and $\left(\frac{2}{e}\right) = 1$, a contradiction.

(ii) Put $e := 4abd - n$. Then $ce = 4abcd - nc = a + b > 0$, so $e \ge 1$. Since $\gcd(ab, n) = 1$ we have $\gcd(n,q) = 1$; $e \equiv -n \pmod q$ gives $\left(\frac{e}{q}\right) = \left(\frac{-1}{q}\right)$; and $e \equiv -n \equiv 3 \pmod 4$. As in (i), $\left(\frac{q}{e}\right) = 1$. A prime dividing e and a (or e and b) divides $4abd - e = n$, which is excluded, so $\gcd(ab, e) = 1$, and $b \equiv -a \pmod e$ gives $\left(\frac{ab}{e}\right) = -1$. Hence ab is even, $8 \mid 4abd$, $e \equiv -n \equiv 7 \pmod 8$ and $\left(\frac{2}{e}\right) = 1$, a contradiction. □

Corollary 2.3 (PROVED)**.** *Let κ be an admissible triple of one of the seven families, let P_κ be the product of those coordinates among a,b,c,d,e,f that do not depend on n (the entries of κ, together with e for I1, c for I4, III and II2), and let t be an odd integer with $\gcd(t, P_\kappa) = 1$. Then t^2 does not lie in the class of κ.*

Proof. Suppose it does and put $n = t^2$. By Lemma 2.1 the coordinates a,b,c,d are positive integers with $4abcd = n(a+b)+c$ (Type I) or $4abcd = nc+a+b$ (Type II). For Type I we need $\gcd(c,n) = 1$: in I2, I3 and I4 the coordinate c divides P_κ, and in I1 the relation $4ad\,c = n + f$ gives $\gcd(c,n) \mid f$. For Type II we need $\gcd(abd, n) = 1$: the constant coordinates divide P_κ, and for the others, in II1 $4ab\,d = n + e$ gives $\gcd(d,n) \mid e$, in II2 $fb = nc + a$ gives $\gcd(b,n) \mid a$, and in II3 $4ad\,b = n + e$ gives $\gcd(b,n) \mid e$. In every case the relevant gcd divides P_κ and is therefore 1, and Lemma 2.2 gives a contradiction. □

2.3. **Rules and exponent coverings.** Fix a family $\mathcal{F}(c_0, b, k)$. For an integer $M = 2^a M'$ with M' odd,

$$n_{s+\ell} \equiv n_s \pmod M \qquad \text{whenever } \operatorname{ord}_{M'}(b) \mid \ell \text{ and } b^s \equiv 0 \pmod{2^a}, \tag{2}$$

since $n_{s+\ell} - n_s = c_0 b^s (b^\ell - 1)$. So for every ET or divisor class C the set $\{s \ge s_1 : n_s \in C\}$ is, for s_1 large, a union of residue classes modulo $\operatorname{ord}_{M'}(b)$. More generally:

Definition 2.4. A *rule* for $\mathcal{F}(c_0, b, k)$ consists of integers $s_0 \geq 0$, $\sigma \geq 1$, $\omega \geq 1$ such that $D := \sigma \log_2 b/\omega$ is an integer, and of $x, y, z \in \mathbb{Q}[Y]$ with

$$4xyz = n(Y)\,(xy + yz + zx), \qquad n(Y) := c_0 b^{s_0} Y^D + k.$$

For $s = s_0 + \sigma t$ ($t \geq 0$) put $Y = 2^{\omega t}$; then $n(Y) = n_s$. The rule *applies* at s if x, y, z take positive integer values there. An *exponent covering* of the family is a finite list of rules covering all but finitely many $s \geq 0$, together with explicit solutions for those finitely many exceptions.

ET and divisor classes are rules with $s_0 = 0$, $\sigma = 1$, $Y = b^s$ and $n(Y) = c_0 Y + k$: their coordinates are polynomials in n. The point of this note is to use rules with $D \geq 2$.

Lemma 2.5 (PROVED)**.** *Let a rule be given, and write each of x, y, z as $R_1(Y)/N$ with $R_1 \in \mathbb{Z}[Y]$ and $N = 2^a N'$, N' odd. There are $t^* \geq 0$ and $T \geq 1$ such that for $t \geq t^*$ whether the rule applies at $s = s_0 + \sigma t$ depends only on $t \bmod T$. One may take T the least common multiple of the orders $\mathrm{ord}_{N'}(2^\omega)$ over the three coordinates, and t^* such that $\omega t^* \geq a$ for all three and all numerators are positive on $[2^{\omega t^*}, \infty)$.*

Proof. If some numerator has non-positive leading coefficient, the rule applies at only finitely many s (take $T = 1$ and t^* beyond them). Otherwise all numerators are eventually positive and t^* exists. For $\omega t \geq a$ we have $R_1(2^{\omega t}) \equiv R_1(0) \pmod{2^a}$, and $R_1(2^{\omega t}) \bmod N'$ is periodic in t with period $\mathrm{ord}_{N'}(2^\omega)$. □

When a rule is named after an ET class or a window, we require in addition that the parameters of that class (resp. $(d_1 + d_2)/m$) be integers; the same argument applies to them, and the exponent classes listed in our tables are those on which the named class genuinely applies.

So the set of exponents at which a rule applies is, up to finitely many s, a finite union of residue classes $s \equiv u \pmod{\sigma T}$. The script of Appendix A computes T, t^* and the exact set of admissible residues $t \bmod T$ for every rule, by evaluating one full period. In profinite terms, these sets are clopen subsets of the exponent space $\widehat{\mathbb{Z}}$, and a finite covering of all large s by them covers all of $\widehat{\mathbb{Z}}$, because the integers $s \geq s_1$ are dense in $\widehat{\mathbb{Z}}$. We use this only in the following elementary form.

Lemma 2.6 (PROVED)**.** *If finitely many sets $E_i = \{s \equiv u_i \pmod{P_i},\ s \geq s_i\}$ cover all sufficiently large s, then for some i we have $u_i \equiv 0 \pmod{P_i}$, so E_i contains every sufficiently large multiple of P_i.*

Proof. Let $P = \operatorname{lcm} P_i$. A large multiple s of P lies in some E_i, so $u_i \equiv s \equiv 0 \pmod{P_i}$. □

2.4. **Unit-square points.**

Theorem 2.7 (PROVED; [15, Thm. 1.4, Cor. 1.5], [17, Lemma 3.1])**.** *Let c_0 be odd, $k \equiv 1 \pmod 8$ and $c_0 + k = 4^J$. For every $L \geq 1$ and $s_1 \geq 0$ there is no finite set of ET classes and divisor classes containing all $n_s = c_0\, 2^s + k$ with $s \equiv 0 \pmod L$, $s \geq s_1$.*

Proof. Suppose $C_1, \ldots, C_r$ are such classes, of moduli $M_1, \ldots, M_r$, and let $M = \operatorname{lcm} M_i = 2^a M'$ with M' odd. For $s \in S := \{s \geq \max(a, s_1) : L \,\mathrm{ord}_{M'}(2) \mid s\}$ we have $n_s \equiv c_0 + k = 4^J \pmod{M'}$ and $n_s \equiv k \pmod{2^a}$, so $\rho := n_s \bmod M$ does not depend on $s \in S$, and $\rho \in C_i$ for some i. If $C_i = D_q$, then $q \mid M'$ and $q \mid \rho \equiv 4^J \pmod q$, impossible. So C_i is the class of some κ. Since $k \equiv 1 \pmod 8$, k is a square in $\mathbb{Z}_2^\times$; choose by the Chinese remainder theorem an odd integer t with $t \equiv 2^J \pmod{M'}$, $t^2 \equiv k \pmod{2^a}$, and $t \equiv 1$ modulo every prime of P_κ not dividing M. Then $t^2 \equiv \rho \pmod M$, so t^2 lies in the class of κ, and $\gcd(t, P_\kappa) = 1$ (the primes of P_κ dividing M' do not divide $t \equiv 2^J$, and t is odd). This contradicts Corollary 2.3. □

The same proof works for the subfamilies $s \equiv 0 \pmod 2$, i.e. for $c_0 \cdot 4^{s'} + k$; so $3 \cdot 2^s + 1$, $15 \cdot 2^s + 1$, $63 \cdot 2^s + 1$, $7 \cdot 4^s + 9$, $15 \cdot 4^s + 1$, $23 \cdot 4^s + 41$, $31 \cdot 4^s + 33$, $7 \cdot 4^s + 57$, $63 \cdot 4^s + 1$, $47 \cdot 4^s + 17$, $55 \cdot 4^s + 9$ admit no finite ET/divisor covering. The limit point behind the proof is $\mu = 0$ in the exponent space $\widehat{\mathbb{Z}}$: there the q-adic limit of n_s is 4^J for every odd q and the 2-adic limit is k, a unit square at every place. By [2, Prop. 1.9] the statement extends to all primitive classes solvable by polynomials (cf. [14, Prop. 3.1]); we do not need this.

The integer points of $\widehat{\mathbb{Z}}$ that are unit-square points are classified as follows. Write $\nu(\ell)$ for the point of $\prod_q \mathbb{Z}_q$ with odd components $c_0 b^\ell + k \in \mathbb{Z}[1/2]$ and 2-component k.

Proposition 2.8 (PROVED; [15, Cor. 1.5], [17, Lemma 3.1], [16, Lemma 1.2])**.** *Let $k \equiv 1 \pmod 8$ and $b \in \{2, 4\}$. If c_0 is odd, then for $\ell \in \mathbb{Z}$ the point $\nu(\ell)$ is a square of a unit at every prime iff $\ell = 0$ and $c_0 + k = 4^J$ for some $J \geq 1$. If $c_0 \equiv 2 \pmod 4$ and $b = 4$, there is no such ℓ.*

Proof. Let $\ell \geq 0$ and $N = c_0 b^\ell + k$. A positive integer that is a square in $\mathbb{Z}_q$ for all odd q is a perfect square (an integer that is not a square is a non-residue modulo infinitely many primes), and it is a unit at all odd q iff it is a power of 2; so $N = 4^J$. For $\ell \geq 1$ and c_0 odd N is odd and > 1; so $\ell = 0$, $c_0 + k = 4^J$. If $\ell = -a < 0$, the odd components are $(c_0 + kb^a)/b^a$; this is a unit square everywhere iff $b^a(c_0 + kb^a)$ is a perfect square with $c_0 + kb^a$ a power of 2; but $c_0 + kb^a$ is odd and > 1. For $c_0 \equiv 2 \pmod 4$, $b = 4$: $N = c_0 b^\ell + k$ is odd for $\ell \geq 1$, and $c_0 + k \equiv 3 \pmod 4$ is not a power of 2; for $\ell = -a < 0$ we would need $c_0 + k4^a$ to be a perfect square, but it is $\equiv 2 \pmod 4$. □

Remark 2.9 (non-integer points; OPEN)**.** Whether the closure of a family can meet the unit squares at a non-integer point of $\widehat{\mathbb{Z}}$ is an exponential local–global question in the style of Skolem's conjecture and the Hasse-type principle of Bertók and Hajdu [1]; we call it (EH) as in [15, §3]. It is open in general. For $4^s + 105$ a finite screen shows that the closure avoids the unit squares ([16, §1.4]: for every residue s mod 360 some prime $q < 3 \cdot 10^6$ with $\mathrm{ord}_q(4) \mid 360$ divides n_s or has $\left(\frac{n_s}{q}\right) = -1$; a certified finite computation). Beyond the unit squares, the sterile set of the ET menu can contain non-square points; none is known on any of the families studied here.

3. Windows with a forced budget

3.1. **The window lemma.** The following is the elementary criterion for $m/N = 1/y + 1/z$ (Rav [8]; we include the proof), specialised to the first step $4/n = 1/x + m/(nx)$.

Lemma 3.1 (window lemma; PROVED)**.** *Let n, x be positive integers with $m := 4x - n > 0$, and put $N := nx$.*

(a) *If d_1, d_2 are positive divisors of N with $m \mid d_1 + d_2$, then*

$$y = \frac{N(d_1 + d_2)}{m\, d_2}, \qquad z = \frac{N(d_1 + d_2)}{m\, d_1}$$

are positive integers and $4/n = 1/x + 1/y + 1/z$. Conversely every solution with this x arises in this way.

(b) *If $\gcd(m, x) = 1$, $d \mid x^2$ and $m \mid x + d$, then $(x,\ n(x+d)/m,\ nx(x+d)/(md))$ is a solution in positive integers.*

Proof. (a) N/d_2 and $(d_1 + d_2)/m$ are integers, so $y \in \mathbb{N}$, and likewise z; and $1/y + 1/z = \frac{m(d_1+d_2)}{N(d_1+d_2)} = \frac{m}{N} = \frac{4}{n} - \frac{1}{x}$. Conversely, if $m/N = 1/y + 1/z$, let $g = \gcd(y, z)$, $y = gy'$, $z = gz'$; then $mgy'z' = N(y' + z')$ and $\gcd(y'z', y' + z') = 1$, so $y'z' \mid N$; with $d_1 := y' \cdot N/(y'z') = N/z'$

and $d_2 := N/y'$ we get $d_1 + d_2 = N(y' + z')/(y'z') = mg$, and $y = N(d_1 + d_2)/(md_2)$, $z = N(d_1+d_2)/(md_1)$. (b) $y \in \mathbb{N}$ since $m \mid x+d$. For z we show $d \mid x(x+d)/m$. Let p be a prime, $\beta = v_p(x)$, $\gamma = v_p(d) \le 2\beta$. If $\gamma > 0$ then $\beta > 0$ and $p \nmid m$, so $v_p(x(x+d)/m) = \beta + v_p(x+d)$, which is $\ge \beta + \gamma \ge \gamma$ if $\gamma < \beta$, $\ge 2\beta \ge \gamma$ if $\gamma = \beta$, and $= 2\beta \ge \gamma$ if $\gamma > \beta$. The identity is $\frac{1}{y} + \frac{1}{z} = \frac{m}{n(x+d)}(1 + \frac{d}{x}) = \frac{m}{nx}$. □

We call m the *window* or *gap* and $x + d$, resp. the pair (d_1, d_2), the *budget*. For a single integer n the lemma is a lottery: one needs divisors of nx in a prescribed class modulo m. On an exponential family it becomes a theorem on whole residue classes of s as soon as the divisors are *forced* by an algebraic identity.

3.2. **Forced budgets.** Let $\mathcal{F}(c_0, b, k)$ be a family and $m \equiv -k \pmod 4$ a positive integer. On a residue class $s = s_0 + \sigma t$ the integer $x = (n_s + m)/4$ is a polynomial in $Y = 2^{\omega t}$ with rational coefficients, and so is $N = nx$. Every factorisation of N in $\mathbb{Q}[Y]$ gives candidate divisors $d_1(Y), d_2(Y)$; if $m \mid d_1(Y) + d_2(Y)$ for one t in a residue class modulo the period of Lemma 2.5, it holds for the whole class, and Lemma 3.1 produces a rule. The algebraic factorisations used below are:

(SG) *Sophie Germain*: $4Y^4 + w^4 = (2Y^2 + 2Yw + w^2)(2Y^2 - 2Yw + w^2)$ and $Y^4 + 4w^4 = (Y^2 + 2Yw + 2w^2)(Y^2 - 2Yw + 2w^2)$. For $x = c_0 4^{j-1}(4^r + w^4)$ with $r = s - j$ odd one puts $Y = 2^{(r-1)/2}$, $4^r = 4Y^4$.

(CY) *cyclotomic*: for odd h, $Y^h + w^h = \prod_{e \mid h} w^{\varphi(2e)} \Phi_{2e}(Y/w)$; for $x = c_0 4^{j-1}(4^r + w^h)$ with $h \mid r$ one puts $Y = 4^{r/h}$.

(BI) *binomial, moving gap*: for constants a, d, the Type II identity $4abcd = cn + a + b$ is equivalent to

$$(4abd - n)(4acd - 1) = n + 4a^2 d, \tag{3}$$

because the left side equals $4ad(4abcd - b - cn) + n$. So every factorisation $n + 4a^2 d = f\, f'$ in $\mathbb{Q}[Y]$ with $b := (f + n)/(4ad)$, $c := (f' + 1)/(4ad)$ integer-valued and positive on a class gives a rule $(x, y, z) = (abd, acdn, bcdn)$. Its gap $4x - n = f$ moves with s.

In (SG) and (CY) the gap m is constant and the budget moves; in (BI) the gap itself is an algebraic factor. The pieces available in (SG), (CY) are the odd primes of c_0, the polynomial $n(Y)$ itself, and the algebraic factors of x; powers of 2 up to $2^{v_2(c_0 4^{j-1})}$ may be added.

Remark 3.2 (no conflict with the odd-square obstruction)**.** A rule with $D \ge 2$ applies on a set of exponents $\{s \equiv u \pmod P\}$. This set of integers n_s is contained in a residue class of n (for instance modulo $2^P - 1$), which by Theorem 2.7 may well be square-like. There is no contradiction: Theorem 2.7 concerns identities whose coordinates are polynomials in n, while a rule is polynomial in the root $Y = ((n - k)/(c_0 b^{s_0}))^{1/D}$ and is only asserted on the thin family. If a member of the class is a square, the rule simply solves it.

3.3. **Specialisation at a unit-square point.** Let $c_0 + k = 4^J$ and consider rules through the neighbourhood $\{s \equiv 0 \pmod L\}$ of $\mu = 0$ for $n_s = c_0\, 2^s + k$. After the substitution $Y = 2^{s/D}$, such s have $Y \equiv 1$ modulo any fixed odd number and $Y \equiv 0$ modulo any fixed power of 2 once L and s are large. This gives a convenient criterion.

Lemma 3.3 (specialisation; PROVED; [21, Lemma 2.1])**.** *Let $R \in \mathbb{Q}[Y]$ and $D \ge 1$. Then $R(2^{s/D}) \in \mathbb{Z}$ for all s in some set $\{s \equiv 0 \pmod L,\ s \ge s_1\}$ $(D \mid L)$ if and only if $R(1) \in \mathbb{Z}[1/2]$ and $R(0) \in \mathbb{Z}_{(2)}$. More precisely, for every prime power p^e: if $R(2^{s/D}) \in p^e \mathbb{Z}_{(p)}$ for all s in such a set, then $R(1) \in p^e \mathbb{Z}_p$ when p is odd and $R(0) \in 2^e \mathbb{Z}_2$ when $p = 2$.*

Proof. Write $R = R_1/N$ with $R_1 \in \mathbb{Z}[Y]$. For odd $p^e \,\|\, N$ we have $2^{s/D} \equiv 1 \pmod{p^e}$ once $\mathrm{ord}_{p^e}(2) \mid s/D$, so $R_1(2^{s/D}) \equiv R_1(1)$; for $2^e \,\|\, N$, $2^{s/D} \equiv 0 \pmod{2^e}$ once $s/D \geq e$, so $R_1(2^{s/D}) \equiv R_1(0)$. This proves "if". For the converse and the refinement, let $s \to 0$ in $\widehat{\mathbb{Z}}$ along the set (e.g. $s = L \cdot \mathrm{ord}_{p^E}(2) \cdot D \cdot j$ with $E \to \infty$ and j large); then $2^{s/D} \to 1$ in $\mathbb{Z}_p$ for odd p and $2^{s/D} \to 0$ in $\mathbb{Z}_2$, R is continuous on $\mathbb{Q}_p$, and $p^e\mathbb{Z}_p$ is closed. □

Thus a rule polynomial in $Y = 2^{s/D}$ is valid near $\mu = 0$ iff its coordinates are positive for large Y and their values at the two specialisations $Y = 1$ (the member $s = 0$, $n = 4^J$) and $Y = 0$ (the 2-adic limit, $n = k$) are $\mathbb{Z}[1/2]$- resp. $\mathbb{Z}_{(2)}$-integral. Neither specialisation is locally obstructed; whether a rule exists is a global question about polynomial identities. For example, the Sophie Germain rule of Theorem 4.1 below specialises at $Y = X = 1/2$ to $x = K = 3(4X^4+1) \mapsto 15/4$, $f_\pm = 2X^2 \pm 2X + 1 \mapsto 5/2,\ 1/2$: the budget $K + d$ with $d = f_+^2 f_-$ becomes $15/4 + 25/8 = 55/8$, divisible by 11 in $\mathbb{Z}[1/2]$. The odd 2-adic valuation of $f_\pm(1/2)$ is what makes this possible (Lemma 5.3).

4. Theorems

4.1. The Proth family $3 \cdot 2^s + 1$.

Theorem 4.1 (PROVED; [19, Thm. C])**.** *For every $s \geq 0$, $4/(3 \cdot 2^s + 1)$ is a sum of three unit fractions.*

Proof. Write $n = 3 \cdot 2^s + 1$. For $s = 0$, $4/4 = 1/2 + 1/3 + 1/6$.

Step 1: the Sophie Germain window (for $4 \mid s$, $s \geq 4$). Put $X = 2^{(s-4)/4}$, so $n = 48X^4 + 1$, and

$$K := \frac{n+11}{4} = 3(4X^4+1) = 3f_+f_-, \qquad f_\pm := 2X^2 \pm 2X + 1.$$

For $(\beta, \gamma) \in \{(2,1), (0,2), (2,0), (1,2)\}$ let $d = f_+^\beta f_-^\gamma$. Then $d \mid K^2$ in $\mathbb{Z}[X]$, and $\gcd(11, K) = 1$ because $4X^4 \equiv -1 \pmod{11}$ has no solution ($-1/4 \equiv 8$ is not a fourth power, indeed not a square, modulo 11). By Lemma 3.1(b) with $x = K$, $m = 11$,

$$x = K, \qquad y = \frac{n(K+d)}{11}, \qquad z = \frac{nK(K+d)}{11\,d}$$

is a solution as soon as $11 \mid K + d$. Since $K + d = f_+f_-(3 + f_+)$, $f_-(3f_+ + f_-)$, $f_+(3f_- + f_+)$, $f_+f_-(3 + f_-)$ in the four cases and $11 \nmid f_+f_-$, the condition $11 \mid K + d$ reads, respectively,

$$\begin{aligned}
(\beta,\gamma) = (2,1) &: X^2 + X + 2 \equiv 0, && X \equiv 4, 6 \pmod{11},\\
(0,2) &: 2X^2 + X + 1 \equiv 0, && X \equiv 2, 3,\\
(2,0) &: 2X^2 - X + 1 \equiv 0, && X \equiv 8, 9,\\
(1,2) &: X^2 - X + 2 \equiv 0, && X \equiv 5, 7.
\end{aligned}$$

So every $X \not\equiv 0, \pm 1 \pmod{11}$ is covered. All of $n, K, f_\pm, d, K + d$ are positive for real X (the discriminant of $f_\pm$ is -4), so the solution is positive. Since 2 is a primitive root modulo 11 and $X = 2^{(s-4)/4}$, $X \equiv 1$ iff $s \equiv 4 \pmod{40}$ and $X \equiv -1$ iff $s \equiv 24 \pmod{40}$. Hence ES holds for all $s \equiv 0 \pmod 4$ with $s \not\equiv 4, 24 \pmod{40}$.

Step 2: classes polynomial in n. Each line is an identity checked by clearing denominators, valid with positive integer coordinates on the stated class:

condition on n	exponents s	(x, y, z)
$7 \mid n$	$s \equiv 1 \pmod 3$	$(2n/7,\ 15n/7,\ 30n)$
$n \equiv 6 \pmod 7$	$s \equiv 2 \pmod 3$	$(2(n+1)/7,\ 2n,\ 2n(n+1)/7)$
$n \equiv 7 \pmod{15}$	$s \equiv 1 \pmod 4$	$((4n+2)/15,\ 4n,\ 4n(2n+1)/15)$
$n \equiv 13 \pmod{15}$	$s \equiv 2 \pmod 4$	$((4n+8)/15,\ 4n,\ 2n(n+2)/15)$
$5 \mid n$	$s \equiv 3 \pmod 4$	$(2n/5,\ 4n/5,\ 4n)$
$61 \mid n$	$s \equiv 24 \pmod{60}$	$(16n/61,\ 326n/61,\ 2608n)$

(The exponent classes follow from $\mathrm{ord}_7(2) = 3$, $\mathrm{ord}_{15}(2) = 4$, $\mathrm{ord}_{61}(2) = 60$ and $3 \cdot 2^{24} \equiv -1 \pmod{61}$.)

Coverage. Let $s \geq 1$. If $3 \nmid s$ or $4 \nmid s$, a line of Step 2 applies. If $12 \mid s$ and $s \not\equiv 4, 24 \pmod{40}$, Step 1 applies. If $12 \mid s$ and $s \equiv 4$ or $24 \pmod{40}$, then $s \equiv 84$ or $24 \pmod{120}$, so $s \equiv 24 \pmod{60}$ and $61 \mid n$. □

Remark 4.2 (structure and history)**.** (i) The Sophie Germain step is the only one that reaches the unit-square point $\mu = 0$: the class $s \equiv 0 \pmod{40}$ ($X \equiv 6$) lies in Step 1. By Theorem 2.7 no finite set of classes polynomial in n could do this. The two classes $X \equiv \pm 1 \pmod{11}$ that Step 1 misses are the neighbourhoods of the integer point $X = \pm 1$, where $48X^4 + 1 = 7^2$ is a square; those members with $12 \nmid s$ are handled by the classes modulo 7 and 15 of Step 2 (e.g. $s = 4$, $n = 49$, by $7 \mid n$), and those with $12 \mid s$ through 61. (No Type I/II family polynomial in X, in particular no rule of the window-11 menu $d = 3^\alpha f_+^\beta f_-^\gamma$, whose integrality depends only on X mod 33, is valid on a class containing $X = \pm 1$, by the odd-square argument at the integer point $48 + 1 = 7^2$ [19, §4.3]; divisor families by 7 are, e.g. the first line of Step 2.)

(ii) A second, independent certificate was found in [20, Thms. 2.2, 2.3]: for $3 \cdot 4^s + 1$ a tree with nine leaves (three windows $m = 11$, two of Sophie Germain type), and for $6 \cdot 4^s + 1$ the class I3$(1, 5, 3)$ for even $s \geq 2$ and D_5 for odd s. It is re-verified by our script. A third mechanism reaches the same point: for $s \equiv 0, 12 \pmod{36}$, $s \geq 36$, with $Y = 2^{s/3}$, the Type II family $a = 80$, $c = 1$, $d = (21Y + 64)/320$, $b = (Y^2 - 3Y + 9)/7$ is valid. Here a, c are constant and the Type II identity $4abcd = cn + a + b$ is equivalent to $cn + a = b(4acd - 1)$; the two factors are $cn + a = 3(Y + 3)(Y^2 - 3Y + 9)$ and $4acd - 1 = 21(Y + 3)$ [22, Prop. 2.1].

(iii) The family contains the primes $3 \cdot 2^s + 1$ for $s = 1, 2, 5, 6, 8, 12, 18, 30, 36, 41, 66, 189, \ldots$. For $s \geq 3$ a member lies in one of Mordell's six square classes modulo 840 exactly when $12 \mid s$; then $n \equiv 529 \pmod{840}$ ($n \equiv 1 \pmod{24}$, $n \equiv 4 \pmod 5$, $n \equiv 4 \pmod 7$). So the content of Theorem 4.1 beyond Mordell's identities is the case $12 \mid s$.

4.2. **The family $63 \cdot 2^s + 1$.**

Lemma 4.3 (PROVED; [22, Thm. 2.3])**.** *Let $s \equiv 0 \pmod{24}$, $s \geq 24$, $Y = 2^{s/3}$, $n = 63Y^3 + 1 = 63 \cdot 2^s + 1$, $a = 5$, $d = 17$ and*

$$b = \frac{63Y^3 + 3Y^2 - 9Y + 28}{340}, \qquad c = \frac{21Y + 64}{340}.$$

Then $(x, y, z) = (abd,\ acdn,\ bcdn)$ is a solution of (1) in positive integers. Its gap is $4x - n = 3(Y^2 - 3Y + 9)$.

Proof. We have $n + 4a^2d = 63Y^3 + 1701 = 63(Y + 3)(Y^2 - 3Y + 9)$. Put $f = 3(Y^2 - 3Y + 9)$ and $f' = 21(Y + 3)$, so $ff' = n + 1700 = n + 4a^2d$. One checks $340b - n = f$ and $340c - 1 = f'$, i.e. $4abd - n = f$ and $4acd - 1 = f'$; so (3) holds and $4abcd = cn + a + b$, which is the Type II identity (Lemma 2.1). Integrality: $340 = 4 \cdot 5 \cdot 17$; $4 \mid Y$ since $s \geq 6$; $\mathrm{ord}_5(2) = 4$ and $\mathrm{ord}_{17}(2) = 8$, so $8 \mid s/3$ gives $Y \equiv 1 \pmod{85}$, and then $21Y + 64 \equiv 85 \equiv 0$ and $63Y^3 + 3Y^2 - 9Y + 28 \equiv 85 \equiv 0 \pmod{85}$, while both are $\equiv 0 \pmod 4$. Positivity is clear. □

The class $s \equiv 0 \pmod{24}$ contains a neighbourhood of the unit-square point $\mu = 0$ ($63 + 1 = 4^3$). By Theorem 2.7 no class polynomial in n reaches it, and by Theorem 5.7 below no rule with constant gap or constant ratio does either; the gap of Lemma 4.3 is the cyclotomic factor $3(Y^2 - 3Y + 9)$ of $Y^3 + 27$. At $s = 0$ the formulas give $c = 85/340$, so the condition $s \geq 24$ is needed.

Theorem 4.4 (PROVED, computer-assisted; [22, Thm. 3.1])**.** *For every $s \geq 0$, $4/(63 \cdot 2^s + 1)$ is a sum of three unit fractions.*

Proof. For $s \equiv 0 \pmod{24}$, $s \geq 24$, use Lemma 4.3. The eighteen ET and divisor classes of Table 2 apply on the stated exponent classes (for the classes of n this follows from (2) and the listed orders; the identities are those of Table 1). They cover every residue s mod 2520 except $s \equiv 0, 840, 1680 \pmod{2520}$, all of which are $\equiv 0 \pmod{24}$ (computer-checked coverage of 2520 residues). The thresholds in the table are at most 5, and every $s \geq 1$ below the threshold of its class lies in another class of the table without threshold (checked). For $s = 0$, $4/64 = 1/32 + 1/64 + 1/64$. □

class	condition on n	exponents
D(5)	$5 \mid n$	$s \equiv 3 \pmod 4$
II2(1, 2, 15)	$n \equiv 7 \ (15)$	$s \equiv 1 \pmod 4$
II2(2, 2, 15)	$n \equiv 13 \ (15)$	$s \equiv 2 \pmod 4$
II2(5, 3, 119)	$n \equiv 57 \ (119)$	$s \equiv 4 \pmod 8$
II2(5, 2, 39)	$n \equiv 34 \ (39)$	$s \equiv 4 \pmod{12}$
II2(1, 5, 19)	$n \equiv 18 \ (19)$	$s \equiv 14 \pmod{18}$
I1(1, 146, 39)	$n \equiv 545 \ (584)$	$s \equiv 2 \pmod 9$, $s \geq 5$
D(11)	$11 \mid n$	$s \equiv 2 \pmod{10}$
II2(3, 1, 11)	$n \equiv 8 \ (11)$	$s \equiv 4 \pmod{10}$
II2(1, 1, 11)	$n \equiv 7 \ (11)$	$s \equiv 6 \pmod{10}$
D(29)	$29 \mid n$	$s \equiv 20 \pmod{28}$
II2(1, 11, 87)	$n \equiv 43 \ (87)$	$s \equiv 24 \pmod{28}$
D(127)	$127 \mid n$	$s \equiv 1 \pmod 7$
II2(1, 32, 127)	$n \equiv 126 \ (127)$	$s \equiv 2 \pmod 7$
I1(1, 86, 23)	$n \equiv 321 \ (344)$	$s \equiv 4 \pmod{14}$, $s \geq 5$
III1(1, 254, 15)	$n \equiv 1001 \ (1016)$	$s \equiv 5 \pmod 7$, $s \geq 3$
I1(1, 1405, 511)	$n \equiv 5109 \ (5620)$	$s \equiv 28 \pmod{140}$, $s \geq 4$
I2(2, 43, 151)	$n \equiv 193 \ (344)$, $n \equiv 54 \ (151)$	$s \equiv 140 \pmod{210}$, $s \geq 3$

TABLE 2. The ET/divisor classes for $63 \cdot 2^s + 1$ (Theorem 4.4). The exponent classes are derived from the identities by the script (one full period); "$s \geq s_1$" marks where the 2-adic part of the class or positivity needs s large.

Remark 4.5. For $s \geq 3$ we have $n_s \equiv 1 \pmod{168}$ and $n_s \equiv 3 \cdot 2^s + 1 \pmod 5$, so n_s lies in a Mordell square class modulo 840 (namely 169) iff $4 \mid s$; Mordell's identities already handle $s \not\equiv 0 \pmod 4$. The essential content of Theorem 4.4 is the neighbourhood $s \equiv 0 \pmod{840}$ of $\mu = 0$. The only squares in the family are $n_0 = 8^2$ and $n_8 = 127^2$: if $63 \cdot 2^s + 1 = v^2$ with $s \geq 1$ then v is odd, $s \geq 2$, and with $v = 2A + 1$ we get $A(A + 1) = 63 \cdot 2^{s-2}$; the odd one of the coprime factors $A, A + 1$ is a divisor δ of 63 and the other is $(63/\delta)2^{s-2}$, and of the six choices of δ only $\delta = 63$, $2^{s-2} = 64$ works. The square 127^2 lies in the divisor class D_{127}, as it must by Lemma 2.2.

4.3. **The family $7\cdot 4^s+9$.** Here $7+9=4^2$, so $\mu=0$ is a unit-square point and Theorem 2.7 applies. For even $s\ge 2$ one has $n\equiv 121 \pmod{840}$, so every member with even $s\ge 2$ is Mordell-hard. The windows used are $m=7\cdot 4^j t-9$ with $(j,t)\in\{(1,1),(1,4),(2,1),(2,6)\}$, i.e. $m\in\{19,103,663\}$. For such a window and $s>j$,

$$x=\frac{n+m}{4}=7\cdot 4^{j-1}(4^{s-j}+t),\qquad N=nx=4^{j-1}\cdot 7\cdot n\cdot(4^{s-j}+t),$$

and $4^{s-j}+t$ is split by (CY) or (SG): for $t=w^h$ and $h\mid s-j$, with $Y=4^{(s-j)/h}$, into $Y+w$ $(h=1)$, $(Y+1)(Y^2-Y+1)$ $(h=3,\ w=1)$, $(Y+1)(Y^2-Y+1)(Y^6-Y^3+1)$ $(h=9,\ w=1)$; for $(j,t)=(1,1)$ and s even, with $Y=2^{(s-2)/2}$, into f_+f_-, $f_\pm=2Y^2\pm 2Y+1$. A rule is a choice of d_1,d_2 among products of these pieces and of 2^i $(i\le 2j-2)$, and it applies by Lemma 3.1(a) whenever $m\mid d_1+d_2$.

Theorem 4.6 (PROVED, computer-assisted; [20, Thm. 2.6])**.** *For every $s\ge 0$, $4/(7\cdot 4^s+9)$ is a sum of three unit fractions.*

Proof. The 22 window rules of Table 3 and the 11 ET/divisor classes of Table 4 apply on the listed exponent classes; their union contains every residue modulo $9180=2^2\cdot 3^3\cdot 5\cdot 17$ (computer-checked), and all thresholds are ≤ 3. For $s\in\{1,2\}$ the rules apply directly, and for $s=0$, $4/16=1/8+1/12+1/24$. □

m	j	type	Y	d_1	d_2	exponents s
19	1	CY	4^{s-1}	1	$(Y+1)$	$6 \pmod 9$
19	1	SG	$2^{(s-2)/2}$	f_-	7	$18,26 \pmod{36}$
19	1	SG	$2^{(s-2)/2}$	f_+	7	$0,8 \pmod{36}$
19	1	SG	$2^{(s-2)/2}$	1	f_-	$8,32 \pmod{36}$
103	1	CY	4^{s-1}	$(Y+4)$	7	$44 \pmod{51}$
103	1	CY	4^{s-1}	1	n	$8 \pmod{51}$
103	1	CY	4^{s-1}	1	$(Y+4)$	$41 \pmod{51}$
103	2	CY	4^{s-2}	1	$2\cdot 7\cdot n$	$17 \pmod{51}$
103	2	CY	4^{s-2}	2^2	$7\cdot n$	$47 \pmod{51}$
103	2	CY	4^{s-2}	2^2	$7\cdot(Y+1)$	$38 \pmod{51}$
103	2	CY	4^{s-2}	2	$(Y+1)$	$32 \pmod{51}$
103	2	CY_3	$4^{(s-2)/3}$	1	$2\cdot 7\cdot n\cdot(Y^2-Y+1)$	$128 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	$2^2\cdot(Y^2-Y+1)$	$7\cdot n$	$137 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	$2\cdot(Y^2-Y+1)$	$7\cdot(Y+1)$	$32,74 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	2^2	$7\cdot(Y^2-Y+1)$	$29 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	$2\cdot(Y+1)$	$7\cdot(Y^2-Y+1)$	$65,98 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	$n\cdot(Y+1)$	$2^2\cdot 7$	$56 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	2	$n\cdot(Y^2-Y+1)$	$11 \pmod{153}$
103	2	CY_3	$4^{(s-2)/3}$	(Y^2-Y+1)	n	$20 \pmod{153}$
103	2	CY_9	$4^{(s-2)/9}$	1	$2\cdot 7\cdot n\cdot(Y+1)$	$146,155 \pmod{459}$
103	2	CY_9	$4^{(s-2)/9}$	1	$7\cdot n\cdot(Y^2-Y+1)$	$128,308 \pmod{459}$
663	2	CY	4^{s-2}	2	$7\cdot(Y+6)$	$10 \pmod{12}$

TABLE 3. Window rules for $7\cdot 4^s+9$: $x=(n+m)/4$, $y=nx(d_1+d_2)/(md_2)$, $z=nx(d_1+d_2)/(md_1)$ (Lemma 3.1(a)); $f_\pm=2Y^2\pm 2Y+1$. CY_h means the cyclotomic splitting of $4^{s-j}+1=Y^h+1$. Each rule is valid for $s\ge 2$ in its class (for $j=1$ also $s=1$ where applicable).

class	condition on n	exponents
D(11)	$11 \mid n$	$s \equiv 2 \pmod 5$
D(541)	$541 \mid n$	$s \equiv 110 \pmod{270}$
I3(1, 5, 3)	$n \equiv 17\ (20)$, $f \mid n^2+20$	$s \equiv 1 \pmod 2$, $s \geq 2$
I3(1, 11, 3)	$n \equiv 41\ (44)$, $f \mid n^2+44$	$s \equiv 4 \pmod 5$, $s \geq 2$
I3(1, 206, 15)	$n \equiv 809\ (824)$, $f \mid n^2+824$	$s \equiv 50 \pmod{102}$, $s \geq 3$
I3(5, 19, 119)	$n \equiv 261\ (380)$, $f \mid n^2+1900$	$s \equiv 14, 50, 86, 122, 158 \pmod{180}$, $s \geq 2$
II2(1, 2, 31)	$n \equiv 23\ (31)$	$s \equiv 3 \pmod 5$
II2(1, 8, 1887)	$n \equiv 1855\ (1887)$	$s \equiv 30 \pmod{36}$
II2(2, 2, 111)	$n \equiv 79\ (111)$	$s \equiv 12 \pmod{18}$
II2(7, 2, 671)	$n \equiv 279\ (671)$	$s \equiv 26 \pmod{30}$
II2(19, 6, 455)	$n \equiv 436\ (455)$	$s \equiv 4 \pmod 6$

TABLE 4. ET/divisor classes for $7 \cdot 4^s + 9$; for I3 the modulus f is the third parameter.

Remark 4.7. The odd exponents are all handled by I3(1, 5, 3); the work is in the even exponents. The rule that reaches $\mu = 0$ is the Sophie Germain window $m = 19$ with $(d_1, d_2) = (f_+, 7)$ on $s \equiv 0, 8 \pmod{36}$: at the specialisation $Y = 1/2$ one has $f_+ = 5/2$ and $7 + 5/2 = 19/2$. By Lemma 5.3 below, only a piece with odd 2-adic valuation at the limit can do this, and $f_\pm(1/2) \in \{5/2, 1/2\}$ are such pieces. The tree that produced the certificate splits first on s mod 36 (the period of the $m = 19$ rule) and then on the primes 17, 5, 3; the certificate was replayed by two independent checkers in [20].

4.4. **The family $4^s + 105$.** Since $105 = 3 \cdot 5 \cdot 7 \equiv 1 \pmod 8$, $n_s = 4^s + 105 \equiv (2^s)^2 \pmod{105}$ and $n_s \equiv 1 \pmod 8$ for $s \geq 2$: every member with $s \geq 2$ lies in one of Mordell's six square classes modulo 840, and n_s is a square modulo every prime factor of the offset. The family has the non-unit square points $4^2 + 105 = 11^2$, $4^3 + 105 = 13^2$, $4^4 + 105 = 19^2$ and $1 + 105 \cdot 4^2 = 41^2$ ($\mu = -2$), and a greedy search for an ET covering stalled near "mixed" points of $\widehat{\mathbb{Z}}$ that imitate different integer squares at different primes [16, §1.4]. One cubic window class closes it.

Lemma 4.8 (PROVED; [18, Lemmas 1.0, 1.0′])**.** *Let $u \equiv 32 \pmod{151}$ be a positive integer, $n = 256u^3 + 105$ and*

$$x = 64(u^3+1), \qquad A = \frac{64u^3 + 2u^2 - 2u + 66}{151}, \qquad y = nA, \qquad z = 32n(u+1)A.$$

Then $A \in \mathbb{N}$ and $4/n = 1/x + 1/y + 1/z$. If $s \equiv 34 \pmod{45}$ then $u := 4^{(s-4)/3}$ is such an integer and $n = 4^s + 105$.

Proof. $4x - n = 151$, so $4/n - 1/x = 151/(nx)$, and $1/y + 1/z = (32u + 33)/(32n(u+1)A)$; equality with $151/(64n(u^3+1))$ is $151A = 2(32u + 33)(u^2 - u + 1)$, which is the definition of A expanded. For $u \equiv 32$, $32u + 33 \equiv 1057 = 7 \cdot 151 \equiv 0 \pmod{151}$, so $A \in \mathbb{N}$. This is Lemma 3.1(a) with $m = 151 = 4^4 - 105$, $N = nx$, $d_1 = 1$ and $d_2 = 32(u+1)$, a divisor of x forced by $u^3 + 1 = (u+1)(u^2 - u + 1)$; equivalently Lemma 3.1(b) with $d = x/(32(u+1)) = 2(u^2 - u + 1)$. For the second statement, $\operatorname{ord}_{151}(4) = 15$, $(s-4)/3 \equiv 10 \pmod{15}$ and $4^{10} \equiv 32 \pmod{151}$. □

Theorem 4.9 (PROVED, computer-assisted; [18, Thm. 1.1])**.** *For every $s \geq 0$, $4/(4^s + 105)$ is a sum of three unit fractions.*

Proof. The thirteen ET/divisor classes of Table 5 cover every residue s mod 180 except $s \equiv 34, 124 \pmod{180}$, i.e. except $s \equiv 34 \pmod{90}$ (computer-checked), and $s \equiv 34 \pmod{90}$ implies $s \equiv 34 \pmod{45}$, where Lemma 4.8 applies. All moduli are odd and all coefficients are non-negative, so the classes apply for every $s \geq 0$. □

class	condition on n	exponents
D(11)	$11 \mid n$	$s \equiv 2 \pmod 5$
II2(1, 1, 11)	$n \equiv 7 \ (11)$	$s \equiv 0 \pmod 5$
II2(1, 3, 11)	$n \equiv 10 \ (11)$	$s \equiv 1 \pmod 5$
D(41)	$41 \mid n$	$s \equiv 8 \pmod{10}$
II2(2, 53, 1271)	$n \equiv 423 \ (1271)$	$s \equiv 9 \pmod{10}$
II2(3, 22, 527)	$n \equiv 262 \ (527)$	$s \equiv 13 \pmod{20}$
II2(1, 6, 143)	$n \equiv 119 \ (143)$	$s \equiv 24 \pmod{30}$
II2(1, 2, 151)	$n \equiv 143 \ (151)$	$s \equiv 14 \pmod{15}$
D(13)	$13 \mid n$	$s \equiv 3 \pmod 6$
D(19)	$19 \mid n$	$s \equiv 4 \pmod 9$
II2(5, 1, 19)	$n \equiv 14 \ (19)$	$s \equiv 1 \pmod 9$
II2(2, 5, 39)	$n \equiv 37 \ (39)$	$s \equiv 5 \pmod 6$
II2(1, 2, 39)	$n \equiv 31 \ (39)$	$s \equiv 1 \pmod 6$

TABLE 5. ET/divisor classes for $4^s + 105$ (Theorem 4.9).

Remark 4.10. (i) The window class respects the odd-square obstruction locally: for $q = 18837001$, a prime with $\mathrm{ord}_q(4) = 45$, one has $\left(\frac{n_s}{q}\right) = -1$ for every $s \equiv 34 \pmod{45}$ [18, review R246]. (ii) Whether ET and divisor classes alone cover $4^s + 105$ is OPEN: no point of the closure of the family is known to be sterile, the greedy search merely stalled (uncovered Haar measure $8.5 \cdot 10^{-8}$ of the exponent space, EVIDENCE). So the theorem bypasses, and does not refute, a possible ET covering. (iii) $4^s + 105$ is not the first all-Mordell-hard family settled: $315 \cdot 2^m + 1 \equiv 1 \pmod{2520}$ for $m \geq 3$ was settled by an ET covering in [15]. The new feature is a class whose divisor $2(u^2 - u + 1)$ moves with n. The architecture — a finite exponent covering leaving one class, closed by an algebraic factorisation along that class — is Izotov's [4, 3].

4.5. The families $4^s + k$.

Theorem 4.11 (PROVED, computer-assisted; [16]). *For $k \in \{9, 57, 81, 129, 153, 177, 201, 225, 249\}$ and every $s \geq 0$, $4/(4^s + k)$ is a sum of three unit fractions.*

Proof. For each k the ET/divisor classes of Table 6 apply on the listed exponent classes for all $s \geq 0$ (for I1(1, 679, 19) the threshold $s \geq 2$ is below the least member $s = 21$ of its class), and their union contains all residues modulo the least common multiple of the periods (2, 30, 2, 2, 6, 6, 2, 24, 2 respectively). □

Remark 4.12 (scope). By the elementary reduction [16, Lemma 1.1], $4^s + k$ with $s \geq 2$ can meet Mordell's square classes only if $k \equiv 9 \pmod{24}$, which holds for all nine k and for $33, 105, 273, 297$ among $k \leq 300$. For $k \in \{9, 81, 129, 201, 249\}$ no member with $s \geq 2$ lies in a Mordell square class, so these cases are Mordell's theorem; $k = 153$ is Mordell's theorem plus the divisor class $13 \mid n$. The genuinely new coverings are $k = 57$, 177, 225. The families $4^s + k$ have no integer unit-square points (Proposition 2.8), so Theorem 2.7 does not obstruct them; the covering is of Sierpiński type. For $k \in \{33, 273, 297\}$ ES for all $4^s + k$ is OPEN: tree searches with ET classes, windows and sums-of-powers windows stall at uncovered measures $\approx 10^{-7}$ near the square points $4^2 + 33 = 7^2$, $4^4 + 273 = 23^2$, $4^3 + 297 = 19^2$ [16, 18, 20, 22].

k	class	condition on n	exponents
9	D(5)	$5 \mid n$	$s \equiv 0 \pmod 2$
	II2(2, 2, 15)	$n \equiv 13 \ (15)$	$s \equiv 1 \pmod 2$
57	II2(2, 2, 15)	$n \equiv 13 \ (15)$	$s \equiv 0 \pmod 2$
	II2(2, 1, 7)	$n \equiv 5 \ (7)$	$s \equiv 1 \pmod 3$
	II2(1, 1, 7)	$n \equiv 3 \ (7)$	$s \equiv 2 \pmod 3$
	D(11)	$11 \mid n$	$s \equiv 3 \pmod 5$
	II2(1, 1, 11)	$n \equiv 7 \ (11)$	$s \equiv 2 \pmod 5$
	II2(1, 1, 31)	$n \equiv 27 \ (31)$	$s \equiv 0 \pmod 5$
	II2(1, 8, 31)	$n \equiv 30 \ (31)$	$s \equiv 1 \pmod 5$
	II2(7, 9, 2015)	$n \equiv 251 \ (2015)$	$s \equiv 9 \pmod{30}$
81	D(5)	$5 \mid n$	$s \equiv 1 \pmod 2$
	II2(1, 2, 15)	$n \equiv 7 \ (15)$	$s \equiv 0 \pmod 2$
129	D(5)	$5 \mid n$	$s \equiv 0 \pmod 2$
	II2(2, 2, 15)	$n \equiv 13 \ (15)$	$s \equiv 1 \pmod 2$
153	II2(1, 2, 15)	$n \equiv 7 \ (15)$	$s \equiv 1 \pmod 2$
	D(7)	$7 \mid n$	$s \equiv 0 \pmod 3$
	II2(1, 1, 7)	$n \equiv 3 \ (7)$	$s \equiv 1 \pmod 3$
	D(13)	$13 \mid n$	$s \equiv 2 \pmod 6$
177	II2(2, 2, 15)	$n \equiv 13 \ (15)$	$s \equiv 0 \pmod 2$
	II2(1, 1, 7)	$n \equiv 3 \ (7)$	$s \equiv 0 \pmod 3$
	II2(1, 2, 7)	$n \equiv 6 \ (7)$	$s \equiv 1 \pmod 3$
	II2(1, 2, 39)	$n \equiv 31 \ (39)$	$s \equiv 5 \pmod 6$
201	D(5)	$5 \mid n$	$s \equiv 1 \pmod 2$
	II2(1, 2, 15)	$n \equiv 7 \ (15)$	$s \equiv 0 \pmod 2$
225	II2(2, 1, 7)	$n \equiv 5 \ (7)$	$s \equiv 1 \pmod 3$
	II2(1, 1, 7)	$n \equiv 3 \ (7)$	$s \equiv 2 \pmod 3$
	II2(1, 2, 39)	$n \equiv 31 \ (39)$	$s \equiv 0 \pmod 6$
	D(17)	$17 \mid n$	$s \equiv 3 \pmod 4$
	II2(101, 5, 2019)	$n \equiv 1918 \ (2019)$	$s \equiv 9 \pmod{24}$
	I1(1, 679, 19)	$n \equiv 2697 \ (2716)$	$s \equiv 21 \pmod{24}$, $s \geq 2$
249	D(5)	$5 \mid n$	$s \equiv 0 \pmod 2$
	II2(2, 2, 15)	$n \equiv 13 \ (15)$	$s \equiv 1 \pmod 2$

TABLE 6. ET/divisor classes for $4^s + k$ (Theorem 4.11).

5. Limits of the method

5.1. **Constant-gap and constant-ratio families at a unit-square point.** Throughout this subsection c_0 is odd, $k \equiv 1 \pmod 8$, $c_0 + k = 4^J$ (so $J \geq 1$, and $c_0 \equiv 7 \pmod 8$ if $J \geq 2$), and $n_s = c_0\, 2^s + k$.

Definition 5.1. A *root family through* $\mu = 0$ consists of $D \geq 1$, a set $C = \{s \equiv 0 \pmod L,\ s \geq s_1\}$ with $D \mid L$, and $x, y, z \in \mathbb{Q}[Y]$ with $4xyz = n(xy + yz + zx)$ identically, where $n = n(Y) = c_0 Y^D + k$, such that x, y, z take positive integer values at $Y = 2^{s/D}$ for every $s \in C$. It has *constant gap* m if $4x - n = m \in \mathbb{Z}$, and *constant ratio* t if $x = tn$ with $t \in \mathbb{N}$ (in both cases up to permuting x, y, z). A constant ratio $t \in \mathbb{Q}_{>0}$ is automatically in $\mathbb{N}$: integrality of tn on C

gives, by Lemma 3.3, $t4^J \in \mathbb{Z}[1/2]$ and $tk \in \mathbb{Z}_{(2)}$, so the denominator of t has neither an odd prime factor nor a factor 2; hence the headline "constant ratio x/n" is no weaker than $t \in \mathbb{N}$.

Every rule of Definition 2.4 whose exponent class contains all large multiples of some P is, after shrinking the class and rescaling Y, a root family through $\mu = 0$ (a shift $Y \mapsto 2^{-r/D}Y$ with $D \mid r$ is absorbed into the coefficients). Rational functions of Y give nothing new: if $R = A/B \in \mathbb{Q}(Y)$ takes integer values at infinitely many $Y = 2^{s/D}$, then the remainder of A modulo B vanishes (cf. the end of the proof of Lemma 5.2). Families involving other exponentials (3^s, ...), coefficients depending on s, or several unrelated roots are *not* covered by what follows. The constant-gap families include all windows of §3 with any splitting of any piece (cyclotomic, Aurifeuillean, Sophie Germain, splittings by fixed primes, sub-products mixing pieces, the piece n itself) and composite m; constant-ratio families are the Type I shapes with x/n fixed.

Lemma 5.2 (reduction to polynomial divisors; PROVED)**.** *Given a constant-gap root family with gap m, put $N = n\,x \in \mathbb{Q}[Y]$. After shrinking C there are $d_1, d_2, d_3 \in \mathbb{Q}[Y]$ with positive leading coefficients, integer-valued and positive on C, such that $d_1d_2d_3 = N$ and $m \mid d_1(Y) + d_2(Y)$ for $Y = 2^{s/D}$, $s \in C$.*

Proof. Note that $m \geq 1$, since $4/n - 1/x = 1/y + 1/z > 0$ on C, and that m is odd, since n is. On C we have $m/N = 1/y + 1/z$. Write $y/z = U/V$ with $U, V \in \mathbb{Z}[Y]$ coprime in $\mathbb{Q}[Y]$ and with positive leading coefficients, and let $R \neq 0$ be their resultant, so that $\gcd(U(Y), V(Y)) \mid R$ for every integer Y. For every prime $p \mid R$, $\min(v_p(U(Y)), v_p(V(Y))) \leq v_p(R)$ is determined by Y modulo $p^{v_p(R)+1}$. Shrinking C so that $Y = 2^{s/D} \equiv 1$ modulo a high power of every odd $p \mid R$ and $Y \equiv 0$ modulo a high power of 2, the integer $h := \gcd(U(Y), V(Y))$ is constant on C. Then $y' := U(Y)/h$ and $z' := V(Y)/h$ are the coprime parts of y, z: $y = gy'$, $z = gz'$ with $g = \gcd(y, z)$. From $mgy'z' = N(y' + z')$ and $\gcd(y'z', y' + z') = 1$ we get $y'z' \mid N(Y)$ and $mg = (N(Y)/(y'z'))(y' + z')$. For an odd prime $p \mid m$ we have $Y \equiv 1 \pmod p$ on (a shrunk) C, so $n \equiv c_0 + k = 4^J$ and $n + m \equiv 4^J \pmod p$, whence $p \nmid N(Y) = n(n + m)/4$; so $m \mid y' + z'$. Put $d_1 = U/h$, $d_2 = V/h \in \mathbb{Q}[Y]$. Finally d_1d_2 divides N in $\mathbb{Q}[Y]$: write $N = Qd_1d_2 + R_0$ with $\deg R_0 < \deg d_1d_2$; then $R_0(Y)/(d_1d_2)(Y) = N(Y)/(d_1d_2)(Y) - Q(Y)$ lies in $\frac{1}{e}\mathbb{Z}$ for a fixed e and tends to 0, so $R_0(Y) = 0$ for infinitely many Y and $R_0 = 0$. Put $d_3 = N/(d_1d_2)$; it is integer-valued on C because $y'z' \mid N(Y)$, and it has positive leading coefficient because N, d_1, d_2 do. □

Write $\rho_i := d_i(1)$. By Lemma 3.3 $\rho_i \in \mathbb{Z}[1/2]$ and $d_i(0) \in \mathbb{Z}_{(2)}$. Every irreducible factor of $N = n(n + m)/4$ with positive leading coefficient is positive on $[0, \infty)$, because n and $n + m$ have no root there; hence $\rho_i > 0$. Write $\rho_i = 2^{e_i}a_i$ with a_i an odd positive integer.

Lemma 5.3 (Jacobi at the limit; PROVED)**.** *In Lemma 5.2, $a_1a_2a_3 = M := 4^J + m$, $m \mid \rho_1 + \rho_2$ in $\mathbb{Z}[1/2]$, and necessarily $m \equiv 3 \pmod 8$ and $e_1 - e_2$ is odd.*

Proof. $\rho_1\rho_2\rho_3 = N(1) = n(1)(n(1) + m)/4 = 4^{J-1}M$, which gives the first claim. For a prime power $p^v \parallel m$ we have $d_1 + d_2 \in p^v\mathbb{Z}_{(p)}$ on C, so $\rho_1 + \rho_2 \in p^v\mathbb{Z}_p$ by Lemma 3.3. Since $n \equiv k \equiv 1 \pmod 4$, $m = 4x - n \equiv 3 \pmod 4$. Every prime $q \mid M$ is odd and satisfies $-m \equiv 4^J \pmod q$, so $\left(\frac{m}{q}\right) = \left(\frac{-1}{q}\right)$ and, by reciprocity ($m \equiv 3 \pmod 4$), $\left(\frac{q}{m}\right) = \left(\frac{m}{q}\right)\left(\frac{-1}{q}\right) = 1$. Hence $\left(\frac{a_i}{m}\right) = 1$ (note $\gcd(M, m) = \gcd(4^J, m) = 1$), and $\rho_1 \equiv -\rho_2$ gives $\left(\frac{2}{m}\right)^{e_1 - e_2} = \left(\frac{-1}{m}\right) = -1$. So $\left(\frac{2}{m}\right) = -1$, i.e. $m \equiv 3 \pmod 8$, and $e_1 - e_2$ is odd. □

This is the window analogue of the odd-square obstruction: every piece is "square-like" at the limit except powers of 2, and only an odd power of 2 can produce the non-residue -1 modulo m.

Lemma 5.4 (2-adic size of the limits; PROVED)**.** *Assume $n = c_0 Y^D + k$ is irreducible in $\mathbb{Q}[Y]$. Write $d_i = \lambda_i X_i n^{\varepsilon_i}$ with $\lambda_i \in \mathbb{Q}$, X_i a monic divisor of x of degree $\delta_i D$, and $\varepsilon_i \in \{0,1\}$ (so $\varepsilon_1+\varepsilon_2+\varepsilon_3 = 1$ and $\delta_1+\delta_2+\delta_3 = 1$). Then each $d_i(0)$ is a 2-adic unit and*

$$e_i = -2\delta_i + 2J\varepsilon_i.$$

Proof. $N(0) = k(k+m)/4$, and $k+m \equiv 4 \pmod 8$ by Lemma 5.3, so $N(0)$ is a 2-adic unit; the $d_i(0) \in \mathbb{Z}_{(2)}$ multiply to it, so each is a unit, and $e_i = v_2(d_i(1)/d_i(0))$, independent of λ_i. For a monic P with roots α in $\overline{\mathbb{Q}}_2$, $P(1)/P(0) = \prod_\alpha (1-\alpha)/(-\alpha)$. The roots of x satisfy $\alpha^D = -(k+m)/c_0$, so $v_2(\alpha) = 2/D > 0$, $v_2(1-\alpha) = 0$, and each contributes $-2/D$; this gives $v_2(X_i(1)/X_i(0)) = -2\delta_i$. For n, $v_2(n(1)/n(0)) = v_2(4^J/k) = 2J$. □

Lemma 5.5 (Capelli at the 2-adic place; PROVED)**.** *If some $\delta_i = \frac{1}{2}$ (i.e. x has a monic divisor of degree exactly $D/2$) and c_0 is fourth-power-free, then $4 \mid D$ and $m = 4c_0w^4 - k$ for an odd integer $w \geq 1$.*

Proof. Up to a constant, $x = Y^D + K$ with $K = (k+m)/c_0 > 0$, $v_2(K) = 2$, and x is reducible. By Capelli's theorem [5, Ch. VI, Thm. 9.1], $-K$ is a p-th power in $\mathbb{Q}$ for a prime $p \mid D$, or $4 \mid D$ and $-K \in -4\mathbb{Q}^4$. The first is impossible for $p = 2$ ($-K < 0$) and for odd p ($p \nmid v_2(K) = 2$). So $K = 4\beta^4$ with $\beta = w/u \in \mathbb{Q}_{>0}$ in lowest terms; as $v_2(\beta) = 0$, w and u are odd, and $(k+m)u^4 = 4c_0w^4$ forces $u^4 \mid c_0$, so $u = 1$. □

Lemma 5.6 (magnitude; PROVED)**.** *Let $e := |e_1 - e_2| \leq E$ and $r := M/m$. Then*

$$m \leq B := \max(2^E 4^J + 2^{2E+2}r^2,\ 2^{E+2}r^2 + 4^J).$$

Proof. By symmetry let $e_1 \geq e_2$, and put $u = 2^e a_1$, $v = a_2$; then $m \mid u+v$ and $u+v > 0$. If $u \geq v$, then $u \geq m/2$, so $a_1 \geq m/2^{e+1}$ and $a_2a_3 = M/a_1 \leq 2^{e+1}r$. Multiplying $u \equiv -v \pmod m$ by a_2a_3 gives $m \mid 2^eM + a_2^2a_3$, and $M \equiv 4^J \pmod m$, so m divides the positive integer $2^e4^J + a_2^2a_3 \leq 2^E4^J + 2^{2E+2}r^2$. If $v > u$, then $a_2 > m/2$, $a_1a_3 < 2r$, and multiplying by a_1a_3 gives $m \mid 2^e a_1^2 a_3 + 4^J \leq 2^{E+2}r^2 + 4^J$. □

Theorem 5.7 (Theorem W; PROVED up to the finite enumeration of Computation 5.8; [22, Thm. W])**.** *Let c_0 be odd and fourth-power-free, $k \equiv 1 \pmod 8$, $c_0 + k = 4^J$, and assume $c_0Y^D + k$ is irreducible in $\mathbb{Q}[Y]$ for every $D \geq 1$. If a constant-gap root family through $\mu = 0$ exists, then there are an odd $w \geq 1$ with $m = 4c_0w^4 - k \in (0, B]$ (B as in Lemma 5.6 with $E = 2J+2$), odd divisors a_1, a_2 with $a_1a_2 \mid M = 4^J + m$, and $e_i = -2\delta_i + 2J\varepsilon_i$ with $2\delta_i \in \{0,1,2\}$, $\delta_1+\delta_2 \leq 1$, $\varepsilon_i \in \{0,1\}$, $\varepsilon_1+\varepsilon_2 \leq 1$, such that $m \mid 2^{e_1}a_1 + 2^{e_2}a_2$ in $\mathbb{Z}[1/2]$.*

Proof. Lemmas 5.2–5.4 give d_i, $\rho_i = 2^{e_i}a_i$ with $e_i = -2\delta_i + 2J\varepsilon_i$. Since $e_i \in \mathbb{Z}$, $2\delta_i \in \mathbb{Z}$; with $0 \leq \delta_i \leq 1$ this gives $2\delta_i \in \{0,1,2\}$. By Lemma 5.3, $e_1 - e_2$ is odd, so some $2\delta_i$ is odd, i.e. $\delta_i = \frac{1}{2}$, and Lemma 5.5 gives $m = 4c_0w^4 - k$ with w odd. As $-2 \leq e_i \leq 2J$, $|e_1 - e_2| \leq 2J+2$, and Lemma 5.6 bounds m. □

Computation 5.8 (COMPUTER-ASSISTED)**.** For each (c_0, k) in $\{(15,1),(31,33),(7,57),(63,1),(47,17),(55,9)\}$ the hypotheses of Theorem 5.7 hold (Capelli: k/c_0 is not a p-th power for odd p and not in $4\mathbb{Q}^4$), and the enumeration over all odd w with $0 < m \leq B(m)$ ($w \leq 3,5,9,5,5,5$ respectively; $B(m)$ decreases in m), all a_1, a_2 and all admissible (e_1, e_2) — without imposing the parity of $e_1 - e_2$ — finds *no* relation (function `thmW` of the script). Positive controls: for $(3,1)$, $(7,9)$, $(23,41)$, $(39,25)$ the same enumeration finds relations exactly at $m = 11$, 19, 51, 131, the

Sophie Germain windows that do reach $\mu = 0$; for $(7, 9)$ the relations include $(\rho_1, \rho_2) = (5/2, 7)$, the specialisation of the rule $(d_1, d_2) = (f_+, 7)$ of Remark 4.7.

Corollary 5.9 (constant ratio; PROVED)**.** *Under the hypotheses of Theorem 5.7 (irreducibility of n suffices), no root family with $x = tn$, $t \in \mathbb{N}$, exists.*

Proof. Then $\varepsilon := 4t - 1 \equiv 3 \pmod 4$ and $\varepsilon/(tn) = 1/y + 1/z$. The proof of Lemma 5.2 applies with m replaced by ε and N by tn (an odd $p \mid \varepsilon$ does not divide t, and $tn \equiv t4^J \pmod p$ near $\mu = 0$), and gives $d_1 d_2 d_3 = tn$ with $\varepsilon \mid d_1 + d_2$. As n is irreducible, $d_i = \lambda_i n^{\varepsilon_i}$ with $\lambda_1\lambda_2\lambda_3 = t$. By Lemma 3.3, $\lambda_i k^{\varepsilon_i} = d_i(0) \in \mathbb{Z}_{(2)}$, so $v_2(\lambda_i) \geq 0$, and $\lambda_i 4^{J\varepsilon_i} = d_i(1) \in \mathbb{Z}[1/2]$, so the odd parts of the λ_i are positive integers dividing t. For an odd prime $q \mid t$, $\varepsilon \equiv -1 \pmod q$, so $\left(\frac{\varepsilon}{q}\right) = \left(\frac{-1}{q}\right)$ and $\left(\frac{q}{\varepsilon}\right) = 1$. If t is odd, all λ_i are odd; if t is even, $\varepsilon \equiv 7 \pmod 8$ and $\left(\frac{2}{\varepsilon}\right) = 1$. In both cases $\left(\frac{\rho_1\rho_2}{\varepsilon}\right) = 1$, contradicting $\rho_1 \equiv -\rho_2 \pmod \varepsilon$ and $\left(\frac{-1}{\varepsilon}\right) = -1$. □

Corollary 5.10 (PROVED, with Computation 5.8)**.** *The families $15 \cdot 2^s + 1$, $31 \cdot 4^s + 33$, $7 \cdot 4^s + 57$, $63 \cdot 2^s + 1$, $63 \cdot 4^s + 1$,*

$47 \cdot 4^s + 17$ and $55 \cdot 4^s + 9$ admit no exponent covering by finitely many ET classes, divisor classes, and rules that are constant-gap or constant-ratio root families (of any degrees D).

Proof. By Lemma 2.6 some rule of a finite covering applies on all large multiples of some P (for the 4^s families: all large even multiples, i.e. all large multiples of $2P$ in the base-2 exponent). It is not an ET or divisor class (proof of Theorem 2.7), and not a constant-gap or constant-ratio root family (Theorem 5.7, Computation 5.8, Corollary 5.9). □

Remark 5.11 (scope and history)**.** (i) Theorem 5.7 is about the *menu*; it does not say that ES fails for any member. It explains why the Sophie Germain mechanism works for $(3, 1)$ and $(7, 9)$ but cannot work for $(15, 1)$ or $(63, 1)$, and *moving-gap* families (§3, (BI)) are not covered: Lemma 4.3 is such a family, so for $63 \cdot 2^s + 1$ (and its subfamily $63 \cdot 4^s + 1$) the obstruction of Corollary 5.10 is real but not an obstruction to ES.unsplit pieces [20, Prop. 2.7, Cor. 2.8] or restricted Sophie Germain menus [19, Props. 4.1, 4.3]; Theorem 5.7 contains them. (iii) An analogue at the non-unit square point of $4^s + 297$ ($4^3 + 297 = 19^2$) holds for windows m with $19 \nmid m$ and for constant ratio; for $19 \mid m$ it is OPEN [22, Prop. 4.1].

5.2. **The family $15 \cdot 2^s + 1$.** Here $15 + 1 = 4^2$: Theorem 2.7 and Corollary 5.10 apply, and no rule reaching $\mu = 0$ is known. Away from $\mu = 0$ ET classes suffice.

Proposition 5.12 (PROVED, computer-assisted; new)**.** *For every $s \geq 1$ with $60 \nmid s$, $4/(15 \cdot 2^s + 1)$ is a sum of three unit fractions.*

Proof. The eleven classes of Table 7 apply on the listed exponent classes for all $s \geq 0$. By hand: $s \not\equiv 0 \pmod 3$ is covered by the two classes modulo 7 (there $n \equiv 3, 5 \pmod 7$); for $3 \mid s$, the classes with 31 cover $s \equiv 1, 2 \pmod 5$ and those with 11, 99 cover $s \equiv 3, 4, 5, 8 \pmod{10}$; what remains is $3 \mid s$ with $s \equiv 0, 9 \pmod{10}$, i.e. $s \equiv 0, 9, 30, 39 \pmod{60}$, and $s \equiv 9 \pmod{12}$ (class 39), $s \equiv 39$ and $s \equiv 30 \pmod{60}$ (the two classes with $287 = 7 \cdot 41$, $\mathrm{ord}_{287}(2) = 60$) leave only $s \equiv 0 \pmod{60}$. The identities and classes are re-checked by the script. □

Members with $s \equiv 0 \pmod 3$ are Mordell-hard ($n \equiv 121 \pmod{840}$ for $s \geq 3$), so Proposition 5.12 is not covered by Mordell's identities; but every class used is polynomial in n, and the remaining set $s \equiv 0 \pmod{60}$ is a neighbourhood of $\mu = 0$ which, by Theorem 2.7, no further class polynomial in n can remove entirely. Corollary 5.10 excludes constant-gap and constant-ratio rules there. The natural next candidates are the moving-gap binomial menus. We treat the menu (BI) of §3 ("II-ad": a, d constant, $b, c \in \mathbb{Q}[Y]$), which produced Lemma 4.3 for $63 \cdot 2^s + 1$.

class	condition on n	exponents
II2$(1,1,7)$	$n \equiv 3 \ (7)$	$s \equiv 1 \pmod 3$
II2$(2,1,7)$	$n \equiv 5 \ (7)$	$s \equiv 2 \pmod 3$
II2$(2,5,39)$	$n \equiv 37 \ (39)$	$s \equiv 9 \pmod{12}$
D(31)	$31 \mid n$	$s \equiv 1 \pmod 5$
II2$(1,8,31)$	$n \equiv 30 \ (31)$	$s \equiv 2 \pmod 5$
II2$(1,3,11)$	$n \equiv 10 \ (11)$	$s \equiv 4 \pmod{10}$
II2$(1,5,99)$	$n \equiv 79 \ (99)$	$s \equiv 8 \pmod{10}$
D(11)	$11 \mid n$	$s \equiv 3 \pmod{10}$
II2$(3,1,11)$	$n \equiv 8 \ (11)$	$s \equiv 5 \pmod{10}$
II2$(1,3,287)$	$n \equiv 275 \ (287)$	$s \equiv 39 \pmod{60}$
II2$(1,24,287)$	$n \equiv 191 \ (287)$	$s \equiv 30 \pmod{60}$

TABLE 7. ET/divisor classes for $15 \cdot 2^s + 1$, $60 \nmid s$ (Proposition 5.12).

Lemma 5.13 (limit conditions for (BI); PROVED; [23, Lemma 2.1])**.** *Suppose a Type II root family through $\mu = 0$ of $n = 15Y^D + 1$ has $(x, y, z) = (abd, acdn, bcdn)$ with $a, d \in \mathbb{N}$ constant and $b, c \in \mathbb{Q}[Y]$ integer-valued and positive on C (the menu (BI) of §3), and put $f = 4abd - n$, $f' = 4acd - 1$, so that $ff' = n + 4a^2d$ by (3). If f and f' are both non-constant, then*

(C1) *$K := 1 + 4a^2d = 15u^p$ for an odd prime $p \mid D$ and an integer $u \geq 3$ with $u \equiv 3 \pmod 4$;*
(C2) *$X := f(1)$ and $X' := f'(1)$ are positive integers with $XX' = 16 + 4a^2d = 15(1 + u^p)$, $X \equiv -16$ and $X' \equiv -1$ modulo q, the odd part of ad.*

If f is constant, the family has constant gap; if f' is constant, c is constant and the family is the ET class II2.

Proof. If f' is constant then so is c, and $b = (cn + a)/f'$ is a polynomial in n: an ET class of type II2, excluded at $\mu = 0$ by Theorem 2.7. If f is constant the gap $4x - n = f$ is constant (Theorem 5.7). Otherwise $n + 4a^2d = 15(Y^D + K/15)$ is reducible, and by Capelli's theorem $-K/15$ is a p-th power for a prime $p \mid D$ or lies in $-4\mathbb{Q}^4$ with $4 \mid D$. The second needs K even, but K is odd; $p = 2$ is impossible as $-K/15 < 0$. So $K/15 = u^p$ with p odd and $u = r/v \in \mathbb{Q}_{>0}$ in lowest terms; $Kv^p = 15r^p$ forces $v^p \mid 15$, so $v = 1$. K is odd, so u is odd; $u = 1$ would give $4a^2d = 14$; so $u \geq 3$. From $K \equiv 1 \pmod 4$ and $u^p \equiv u \pmod 8$ we get $-u \equiv 1 \pmod 4$. For (C2): f, f' are positive on C, so they have positive leading coefficients, and since $n + 4a^2d$ has no root in $[0, \infty)$ neither factor has; so $X, X' > 0$. By Lemma 3.3 applied to $b = (f + n)/(4ad)$ and $c = (f' + 1)/(4ad)$, for every odd prime ℓ, $b(1), c(1) \in \mathbb{Z}_{(\ell)}$, hence $X = 4ad\, b(1) - 16$ and $X' = 4ad\, c(1) - 1$ lie in $\mathbb{Z}_{(\ell)}$, with $\ell^{v_\ell(ad)} \mid X + 16$ and $\ell^{v_\ell(ad)} \mid X' + 1$. At 2: $f(0) = 4ad\, b(0) - 1 \in \mathbb{Z}_{(2)}$, and $f = \kappa P$ with P monic whose roots α are roots of $n + 4a^2d$, hence 2-adic units ($\alpha^D = -K/15$); so $v_2(P(1)) \geq 0 = v_2(P(0))$ and $v_2(X) \geq v_2(f(0)) \geq 0$. Likewise for X'. Finally $XX' = (n + 4a^2d)(1) = 16 + 4a^2d = 15 + K$. □

Lemma 5.14 (PROVED + exhaustive check; cf. [23, Lemma 2.2′])**.** *The equation $a(m_1m_2d - 4) = m_1 + 16m_2$ has exactly 39 solutions in positive integers. The proof gives a priori $a \leq 81$, $m_1 \leq 100$, $m_1d \leq 120$ (the solutions have $a \leq 81$, $m_1 \leq 36$, $m_1d \leq 40$);31 of them have a and d odd, and among these $15 \mid 4a^2d + 1$ only for $(a, d) = (1, 11)$, where $(4a^2d + 1)/15 = 3$.*

Proof. The equation is equivalent to $m_2(aP - 16) = m_1 + 4a$ with $P := m_1d$, so $aP > 16$. If $m_2 = 1$: $a(P - 4) = m_1 + 16 \leq P + 16$, so $P \geq 5$ and $a \leq (P + 16)/(P - 4) \leq 21$; also $m_1(ad - 1) = 4a + 16$, so $ad \geq 2$, $m_1 \leq 4a + 16 \leq 100$ and $P = 4 + (m_1 + 16)/a \leq 120$. If $m_2 \geq 2$: $2(aP - 16) \leq m_1 + 4a \leq P + 4a$, i.e. $(2a - 1)P \leq 32 + 4a$. For $P \geq 5$ this gives $a \leq 6$ and $P \leq 36$.

For $P \le 4$ (so $m_1, d \le 4$) we use $aP > 16$: $P = 1$: $m_2 = (1+4a)/(a-16) = 4 + 65/(a-16)$, so $a - 16 \mid 65$ and $a \le 81$; $P = 2$: $m_2 = 2 + (m_1 + 32)/(2a - 16)$, so $2a - 16 \le 34$, $a \le 25$; $P = 3$: $3m_2 = 4 + (3m_1 + 64)/(3a - 16)$, so $3a - 16 \le 73$, $a \le 29$; $P = 4$: $m_2 = 1 + (m_1 + 16)/(4a - 16)$, so $4a - 16 \le 20$, $a \le 9$. In all cases $a \le 81$, $m_1 \le 100$, $P \le 120$, and m_2 is determined; the script enumerates this range. □

Theorem 5.15 (PROVED; [23, Thm. 2.2])**.** *Every family as in Lemma 5.13 with f, f' non-constant has $u \equiv 7 \pmod 8$; equivalently, a or d is even. In particular the branch $u \equiv 3 \pmod 8$ is empty, for every root degree D and every factorisation of $n + 4a^2 d$.*

Proof. $4a^2 d = 15u^p - 1 \equiv -u - 1 \pmod 8$. If $u \equiv 3 \pmod 8$ this is $\equiv 4 \pmod 8$, so a and d are odd and $q = ad$. By (C2), $X = m_1 ad - 16$ and $X' = m_2 ad - 1$ with integers $m_1, m_2 \ge 1$, and $(m_1 ad - 16)(m_2 ad - 1) = 16 + 4a^2 d$; dividing by ad after expanding gives $a(m_1 m_2 d - 4) = m_1 + 16 m_2$. By Lemma 5.14, $15 \mid 4a^2 d + 1$ forces $(a, d) = (1, 11)$ and $K/15 = 3$, which is not a p-th power with $p \ge 3$. This contradicts (C1). □

Remark 5.16 (the integrality hypothesis)**.** The hypothesis that b and c themselves are integer-valued is part of the menu and is not implied by integrality of x, y, z: the substitution $(a, b, c, d) \mapsto (a/h, b/h, c/h, h^2 d)$ preserves x, y, z, and applied with $h = 5$ to Lemma 4.3 it gives integral constants $a = 1$, $d = 425$ but $b(1) = c(1) = 1/20$. Theorem 5.15 is therefore a statement about the menu (BI) as defined, not about all Type II families with two constant parameters.

Remark 5.17 (what remains open for $15 \cdot 2^s + 1$)**.** (i) The branch $u \equiv 7 \pmod 8$ of (BI) is OPEN. Writing $4a^2 d = 2^E a_1^2 d_1$ ($a_1 d_1$ odd), $E \ge 3$, a solution needs $15u^p - 1$ to have an unusually large part $2^E a_1^2$; if $a_1 \le u^{p/2 - 1 - \epsilon}$ this forces $|15u^p - 1|_2 \le u^{-1-\epsilon}$, a 2-adic approximation of $15^{-1/p}$ by integers u, which by Ridout's theorem has only finitely many (ineffective) solutions for each p,while the sub-branch with large a_1 is of a different, superelliptic type [23, §2.4]. A search found no solution of (C1)+(C2) for $p = 3$, $u \le 2 \cdot 10^4$, and correspondingly smaller ranges up to $p = 23$ (EVIDENCE); the same search recovers the data $(u, p, a, d, X, X') = (3, 3, 5, 17, 21, 84)$ of Lemma 4.3 for $c_0 = 63$. (ii) The other binomial menus (Type II with a, c or c, d constant, Type I with a, c constant) reduce in the same way to limit conditions; for $(15, 1)$ no family was found for small parameters (EVIDENCE, the same searches that find the cubic family of Remark 4.2(ii) for $3 \cdot 2^s + 1$ and Lemma 4.3 for $63 \cdot 2^s + 1$). Searches for one-constant moving-gap families (II with only d constant) also found nothing; for even D such families need $\deg a \ge 2$ [23, Lemma 3.1]. (iii) So ES for $15 \cdot 2^s + 1$, $60 \mid s$, is OPEN; it is the smallest family with a unit-square point for which no rule through $\mu = 0$ is known.

5.3. **Open problems.**

Problem 1. Prove ES for every $15 \cdot 2^s + 1$ with $60 \mid s$. A natural strategy is to find a rule through $\mu = 0$ (valid on all large multiples of some L) and to cover the remaining multiples of 60 by further classes; a rule through $\mu = 0$ would, by Theorem 5.15, have to be a (BI) family on the branch $u \equiv 7 \pmod 8$ or come from a different menu. Conversely, a proof that every rule polynomial in a root $2^{s/D}$ misses $\mu = 0$ would need a sterility theorem for moving-gap families beyond Theorem 5.7; it would not show that ES fails.

Problem 2. Decide $4^s + 33$, $4^s + 273$, $4^s + 297$ and the unit-square families $31 \cdot 4^s + 33$, $7 \cdot 4^s + 57$, $47 \cdot 4^s + 17$, $55 \cdot 4^s + 9$, $23 \cdot 4^s + 41$ (for the last a window reaches $\mu = 0$ but the 2-adic line through it is not covered [20, §2]).

Problem 3. Is the existence of a finite ET/divisor covering of $c_0 \cdot 4^s + k$ ($4 \nmid c_0$, $k \equiv 1 \pmod 8$) equivalent to $c_0 + k \ne 4^J$? This is the conjunction of the exponential local–global statement

(EH) of Remark 2.9 with the absence of sterile non-square points (CONJECTURE E of the notebook [17]).

Problem 4. Is ES for all 2^a3^b+1 provable by a finite certificate? The two $\{2,3\}$-unit-square boundary points are harmless: near the boundary point $(a,b)=(0,1)$, where $n=48X^4+1$, the Sophie Germain window of Theorem 4.1 applies because X tends to $1/2\equiv 6 \pmod{11}$ (it applies whenever $X\not\equiv 0,\pm 1 \pmod{11}$, not to the whole slice $X=2^t3^{t'}$; e.g. $X=3^{10}$ is not covered by the rules of Theorem 4.1); near $(3,0)$ the cubic factorisation $8X^3+1=(2X+1)(4X^2-2X+1)$ provides descent to a smaller Pierpont number

, but a covering of the remaining two-dimensional exponent space was not found [23, §1].

Appendix A. The verification script

The script `scripts/c1_expfam_verify.py` (about 500 lines, sympy and the Python standard library only; run time a few seconds) re-verifies every certificate of this note from the data printed here. For every family it builds each rule as explicit polynomials $x,y,z\in\mathbb{Q}[Y]$ (ET classes from Table 1; windows from m,d_1,d_2 by Lemma 3.1; the classes of Theorem 4.1, Lemmas 4.3, 4.8 verbatim) and then:

(1) checks the identity $4xyz=n(xy+yz+zx)$ in $\mathbb{Q}[Y]$ symbolically;
(2) *derives* the exact set of exponent residues on which the rule applies, by Lemma 2.5: one full period of $R_1(2^{\omega t})$ mod N for each coordinate, beyond the 2-adic and positivity threshold t^* (positivity: no real root of a numerator in $[2^{\omega t^*},\infty)$, by exact Sturm counting, and positive leading coefficient);
(3) checks that the derived classes cover every residue modulo their least common multiple (excluding $s\equiv 0 \pmod{60}$ for $15\cdot 2^s+1$), and that every s below the largest threshold is solved by a rule evaluated at that s, or else by an exhaustive divisor search (needed only for $s=0$ in four families);
(4) builds the solution from the covering rule and checks it exactly for every $s\le 1000$ (every $s\le 600$ for the nine 4^s+k);
(5) runs the finite enumeration of Computation 5.8 with its positive controls, the irreducibility hypothesis of Theorem 5.7 (Capelli), and the enumeration of Lemma 5.14.

The exponent tables of this note were printed from the classes derived in step (2) (helper `scripts/c1_tables.py`); a transcript is in `data/c1/verify_tables.txt`. The second certificate for $3\cdot 2^m+1$ of Remark 4.2(ii) is also re-verified. The original certificates were checked, in addition, by the authors' checkers and by two independent reviewers' programs each (for example `review_r251_proth3.py` and `review_r252_proth3.py` for Theorem 4.1, `review_r257_cert63.py` and `review_r258_cert63.py` for Theorem 4.4, `review_r253_cert.py` and `review_r254_cert.py` for Theorem 4.6, `review_r245_cert.py` and `review_r246_cert105.py` for Theorem 4.9, `review_r235_cert.py` and `review_r236_cert.py` for Theorem 4.11, and `review_r257_thmW.py`, `review_r258_thmW.py` for Computation 5.8); the certificate of Proposition 5.12 is new and has been checked by the script above and by an independent reviewer's program (`review_r291_tables.py`).

Acknowledgements and provenance

This is an internal draft consolidating reviewed entries (J)22, (J)62, (J)69, (J)74, (J)75, (J)80 and (J)102 of the project ledger `DISCOVERIES.md`, with the repairs recorded in the corresponding hostile reviews. No external referee has checked it.

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[14] Companion note in this repository, *Finite congruence coverings for the Erdős–Straus equation and sterile profinite points*, `paper/es-coverings-note.tex`.

[15] Companion note in this repository, `EXPONENT_COVERING.md` (ledger (J)22); reviewed in `reviews/exponent-covering-review.md`, `reviews/exponent-covering-review-2.md`.

[16] Companion note in this repository, `SIGN_FIELDS.md` (ledger (J)62); reviewed in `reviews/sign-fields-review.md`, `-review-2.md`.

[17] Companion note in this repository, `EXP_NEAR_SQUARE.md` (ledger (J)68).

[18] Companion note in this repository, `SYNTHESIS_SPARKS6.md` (ledger (J)69); reviewed in `reviews/synthesis-six-review.md`, `-review-2.md`.

[19] Companion note in this repository, `MENU_STERILITY.md` (ledger (J)74); reviewed in `reviews/menu-sterility-review.md`, `-review-2.md`.

[20] Companion note in this repository, `EXP_105.md` (ledger (J)75); reviewed in `reviews/exp-105-review.md`, `-review-2.md`.

[21] Companion note in this repository, `MENU_LADDER.md` (ledger (J)76); reviewed in `reviews/menu-ladder-review.md`.

[22] Companion note in this repository, `EXP_OPEN.md` (ledger (J)80); reviewed in `reviews/exp-open-review.md`, `-review-2.md`.

[23] Companion note in this repository, `EXP_15.md` (ledger (J)102); reviewed in `reviews/exp-fifteen-review.md`.

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